Who Could Have Rational Expectations?

Bounds Across People and Information Sets

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Abstract

Tests of rational expectations ask whether the hypothesis can be rejected. This paper asks how much of a population could still be rationalized as satisfying it. In a mixture of a rational component and an unrestricted alternative, the mixing weight is bounded above but never below. The upper bound has a closed form for binary outcomes. In the Health and Retirement Study, full rational expectations is rejected decisively, yet at most 97.76 percent can be rationalized given the reported survival belief, and 93.50 percent once beliefs must also be efficient with respect to own sex, age, health and smoking. Testing whether that share is below one is exact in finite samples when observations within a cell are independent, and measuring how far below is not. The bound moves with the information asserted, the outcome forecast, and the population studied. On the same respondents, beliefs about working past sixty-two leave 11.38 percent non-rationalizable where survival beliefs leave 0.32.

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1 Introduction

Empirical work on expectations asks a hypothesis-testing question. Given data on subjective beliefs and subsequent realizations, can rational expectations be rejected? The answer is usually yes. Household forecast errors are correlated with characteristics the respondent knew when making the forecast (Souleles, 2004). Individual professional forecasters overreact to news, while consensus forecasts underreact (Coibion and Gorodnichenko, 2015; Bordalo et al., 2020). Elicited survival probabilities depart from realized mortality in ways that differ with smoking and sex (Hurd, 2009), and rational expectations about earnings is rejected for workers without a college degree and not for college graduates (D'Haultfœuille et al., 2021).

The data behind these tests are simple. For each person a survey records a stated probability that an event will happen and, later, whether it happened, when follow-up is long enough to tell. When the event is binary the person's belief is a single number, the subjective probability, and the survey records it with rounding. Rational expectations in the sense of Muth (1961) implies that this probability equals the objective probability of the event given what the person knows. This paper studies that one observable implication and makes no assumption about preferences or about how the person came to know what they know. Because a person knows their own belief, the implication has content even when nothing else about the person is used. Among people who hold the same belief, the event must occur at that rate. Naming characteristics the person surely knows, such as their own age or whether they smoke, imposes the same restriction within finer groups. The resulting condition is calibration conditional on the named characteristics, and the paper calls a population rationalizable when it could satisfy that condition.

The list of characteristics is an assumption of the analyst, and people are not assumed to choose it. The paper does not model how people acquire information. It requires only that the listed characteristics be part of what the person knows, so a shorter list is a weaker assumption. Only characteristics fixed before the report and known to the person qualify, and knowing a characteristic does not mean the person used it. In the application the list starts empty and adds in turn the respondent's own sex, age group, self-rated health and smoking. These lists are alternative assumptions about the same moment, and they do not describe people learning over time.

As a statistical matter this is a problem about probabilistic forecasts. The contribution is econometric. The paper identifies the largest share of the joint distribution of forecasts and outcomes that a calibrated component can account for when the rest is left unrestricted, and it extends that bound to interval beliefs and censored outcomes with inference and a monotonicity result in the information asserted. It offers no theory of how people acquire information.

A rejection is a statement about a population, and populations are heterogeneous. That some people cannot be calibrated does not tell us how many. This paper asks how large a share of a population could still be calibrated in this sense. Suppose some share of the population satisfies the restriction, and the rest behaves in a way we decline to restrict at all. What do beliefs and realizations reveal about that share?

The answer is asymmetric in a way that organizes the paper. The share is not point identified.

Its identified set runs from zero to an upper bound, and the lower end is zero in every dataset, because an unrestricted alternative can always mimic rational behaviour. Only the upper bound carries information. We call it the maximum rationalizable share, and its complement is a minimum share of the population that no rational-expectations model can accommodate. The bound is the largest share whose beliefs and outcomes admit a rational-expectations representation. Which individuals actually used their information rationally is not identified.

Identification comes from accounting within cells. Condition on a cell defined by the belief and by any characteristics the analyst asserts the agent knows. Inside that cell the rational component must have mean equal to the belief, while the remainder may have any mean at all. That restriction alone caps the rational mass in the cell, in closed form, and the population bound is the average of those caps. The cap equals one only where the realized rate in the cell coincides with the belief, and it falls away linearly on either side, steeply when the belief is near zero or one. Reported probabilities are coarse, so beliefs enter as respondent-specific intervals instead of points. Realizations are sometimes censored, so unresolved outcomes are carried as mass that may be allocated in whichever way most favours rationality. And because the cap is concave in the realized rate, asserting that agents know more weakly lowers the bound, so bounds computed across nested information sets are ordered in the population, not by construction of the estimator alone.

Testing whether the bound equals one is far easier than estimating how far below one it lies. Full rational expectations is feasible if and only if every cell's realized rate is compatible with some belief the cell admits, so refuting it requires that a single cell fail, and each cell contributes a pair of one-sided hypotheses about one binomial proportion. Under independent sampling that test is exact in finite samples, and it needs no linear program, no bootstrap and no estimated weights. Bounding the magnitude requires aggregating evidence across every cell and every cell weight at once, and the resulting functional is directionally but not fully differentiable, a class in which the standard bootstrap is consistent only where the functional is fully differentiable (Fang and Santos, 2019). The contrast explains why the literature reports rejections of rational expectations far more often than it reports how much of a population those rejections implicate.

We apply the framework to personal survival beliefs in the Health and Retirement Study, where respondents report the percent chance of reaching an individually tailored target age and the realization is later observed for the same person. Full rational expectations is rejected, and the rejection survives two treatments that relax the within-cell independence the binomial test assumes. The violations run in both directions and sit at both tails of the belief distribution. Respondents whose reports admit no more than a three to five percent chance reach their target age about thirteen percent of the time. Respondents whose reports admit nothing below ninety-five percent reach it between twenty-seven and thirty-six percent of the time, and even counting every respondent whose outcome is still undetermined as a survivor they fall short of the lowest belief their own reports admit.

The identified magnitude is positive and small. Conditioning on the reported belief alone, at most 97.76 percent of the weighted population reporting on a grid finer than the focal one could be

rationalized, so at least 1.82 percent of them could not, at ninety-five percent confidence. Extending the statement to everyone eligible, by treating the respondents whose reports admit no point belief in the way most favourable to rationality, gives a floor of 1.52 percent. Requiring beliefs to be efficient with respect to the respondent's own sex, age, self-rated health and smoking lowers the maximum rationalizable share to 93.50 percent and raises the certified floor to 4.90 percent. The floor nearly triples, from about two people in a hundred to about five. Two checks show that the decline is not an artifact of cutting the data more finely. Adding a variable does not lower the bound automatically. The respondent's own sex leaves it unchanged where the ladder adds it. And a population that is rational by construction, evaluated on the same cells, produces about a quarter of that rise, which is measured and reported. Each rung therefore detects a conditional discrepancy between beliefs and outcomes that coarser cells conceal, and not the predictive power of the variable added.

The bound is also not a fixed attribute of the population. Applying the same architecture to beliefs about working full time after sixty-two leaves 11.38 percent of the same respondents non-rationalizable where their survival beliefs leave 0.32, and the survival bound itself varies by an order of magnitude across the subsamples that different questions define. What is being forecast, the information asserted and the population studied all move the answer.

The paper changes the estimand relative to the closest existing work. Tests of rationalizability deliver a population-level verdict on whether beliefs and realizations can be reconciled at all (D'Haultfœuille et al., 2021). Switching models estimate time-varying shares of agents who use each of a few fully specified forecasting rules, from individual survey responses (Branch, 2004) or from aggregate inflation series (Cornea-Madeira et al., 2019), and their forward-looking rule is a forecast that the authors distinguish from rational expectations. In choice data, Kitamura and Stoye (2018) test whether a population of utility maximizers can generate observed choice frequencies, and Caliari and Petri (2024) bound from below the fraction of decision makers who must be rational. The closest object in econometrics is a sharp bound on a population share obtained by optimizing over every latent structure the data admit, as in the bound of Torgovitsky (2019a) on the share of unemployment persistence due to state dependence, and the complement of the share is a mixture index of fit in the sense of Rudas et al. (1994). This paper supplies a bound of that kind for expectations data. The bound is sharp, nonparametric and individual-level, the non-rational component is left unrestricted, and the monotonicity result makes the sequence of bounds across information sets an object with content. Sharpness is a property of the population bound. The empirical implementation relaxes it, letting cells pool respondents whose belief intervals differ, because the partition that delivers the sharp bound produces cells too small to estimate. The framework also extends to several outcomes at once, with a characterization of when requiring rationalizability in every domain simultaneously restricts more than the domain-by-domain bounds do.

Section 2 places the paper in the literature and Section 3 develops identification. Sections 4 through 6 contrast the two inference problems and report the data and the results. Proofs, supporting material and the multidomain extension are in the appendices.

2 Related Literature

Panel tests on the individual forecasts of professional forecasters avoid aggregation bias and allow errors to be correlated across forecasters, and with those corrections they do not reject the rationality of price forecasts (Keane and Runkle, 1990). Coibion and Gorodnichenko (2015) regress consensus forecast errors on consensus forecast revisions to measure information rigidity, and Bordalo et al. (2020) run the same regression on individual forecasters and find overreaction for individuals alongside underreaction in the consensus. In household data, forecast errors are correlated with demographic characteristics the respondent knows when forecasting (Souleles, 2004). Among Finnish men, higher measured IQ predicts smaller inflation forecast errors and consumption plans that respond to expected inflation as the consumer Euler equation prescribes (D'Acunto et al., 2023). A parallel strand asks whether apparent irrationality can be rationalized by asymmetric loss (Elliott et al., 2005), a defence that is weaker for probability reports about binary personal events than for point forecasts of continuous variables. This work asks whether the hypothesis can be rejected, and it does not report what fraction of the population a rejection implicates.

The measurement tradition this paper draws on elicits probabilistic expectations in household surveys (Manski, 2004). The resulting probabilities predict later behaviour and outcomes where respondents hold private information, as with survival and retirement (Hurd, 2009). In the Health and Retirement Study, average survival probabilities to age seventy-five are close to life-table rates while those to eighty-five exceed them, and individual reports predict mortality after controlling for self-rated health (Hurd and McGarry, 2002). More recent work compares elicited beliefs with individual outcomes. The job-finding probabilities that unemployed workers report predict the employment outcomes they later report, although they are optimistic and are not revised down as unemployment continues (Mueller et al., 2021). Survey inflation forecasts linked to Danish administrative records covary with household income and health shocks far more than realized inflation does (Taubinsky et al., 2026). The estimand here is a population share.

The econometric ancestor of that share is contaminated sampling, in which a known upper bound on the probability of contamination yields bounds on features of the uncontaminated distribution (Horowitz and Manski, 1995). The logic is inverted here, with the restriction fixed and the mixing weight bounded. That inversion is the mixture index of fit of Rudas et al. (1994), the smallest share of a population that must be set aside for a model of a contingency table to fit exactly while the remainder is left unrestricted. The minimum non-rationalizable share is an index of that kind. What this paper adds is the restriction it applies, the one rational expectations places on probabilistic beliefs. It also handles beliefs observed only as intervals and outcomes that are partly censored, and it supplies an exact test and inference for the resulting bound. Without further restrictions the components and weights of a finite mixture are not identified. An exclusion restriction yields a sharp identified set, and point identification of a two-component mixture needs further conditions such as dominance of one component in each tail (Henry et al., 2014). Nonparametric identification of the components becomes possible with three or more conditionally independent measurements per unit (Bonhomme et al., 2016), and grouped fixed-effects estimators

classify units consistently as the number of periods grows and when groups are well separated (Bonhomme and Manresa, 2015; Bonhomme et al., 2022). None of these conditions holds here. Each respondent contributes one belief and one realization, and the alternative population is unrestricted, so it can reproduce any rational cell exactly. That is why the share has only an upper bound and why individual membership is not identified. Repeated belief and realization pairs per person, combined with a restriction on the alternative population, would bring the question of who the non-rationalizable respondents are within reach of these methods.

The bound belongs to a literature that computes sharp identified sets by optimizing over every latent structure consistent with the data. Mogstad et al. (2018) bound policy-relevant treatment parameters by searching over the marginal treatment response functions that reproduce a set of instrumental-variable estimands, and the endpoints of the bound solve two convex programs that are linear when the admissible set is polyhedral. Torgovitsky (2019b) characterizes sharp identified sets by asking whether some distribution of unobservables can generate the observed distribution, and Torgovitsky (2019a) uses linear programming over the distribution of latent potential outcome sequences to bound the share of unemployment persistence due to state dependence. The maximum rationalizable share is an object of the same kind. Its program separates across cells and has a closed-form solution, and this is what permits the exact finite-sample test of full rational expectations below. Restrictions that link cells, such as monotonicity of the alternative population in the reported probability, would remove the closed form, and the linear-programming formulations of these papers are the route for imposing them.

What agents know when they form expectations has a longer history in work that separates heterogeneity agents can forecast from uncertainty they cannot (Cunha et al., 2005). There a candidate information set is valid when the part of realized outcomes it leaves unexplained does not predict the agent's choices, so the content of the information set is inferred from choices and realizations together. The condition used here has the same structure with the elicited probability in place of the choice, since relative to the asserted information the realization minus the report must have mean zero. Cunha et al. (2005) also find that person-specific information sets are generally not identified and only their distribution is, which parallels the result here that a population share can be bounded while individual membership cannot be recovered.

Two literatures come closest on expectations themselves. The first characterizes when a distribution of beliefs and a distribution of realizations can be rationalized by rational expectations at all, showing that rationalizability is equivalent to the realizations being a mean-preserving spread of the beliefs, and tests that hypothesis using only marginal distributions from possibly unmatched datasets (D'Haultfœuille et al., 2021). That test delivers a population-level verdict, and for a binary outcome it reduces to comparing the mean belief with the mean realization. When beliefs and realizations can be matched person by person, as here, a stronger restriction is available and it supports a share. An earlier partial-identification treatment derives bounds on features of the realized distribution under a best-case rationality hypothesis with categorical predictions (Das et al., 1999), which anticipates the logic without producing the mixture object. The second literature estimates

time-varying shares of agents choosing among a few fully specified forecasting rules in a switching model, with a forecast from an estimated vector autoregression standing in for the forward-looking rule. Branch (2004) estimates the shares of agents using that forecast, adaptive expectations and naive expectations from individual responses to the Michigan survey, and Cornea-Madeira et al. (2019) estimate the share of forward-looking fundamentalists against naive forecasters from aggregate United States inflation. Both papers distinguish the forward-looking rule from rational expectations, and the shares they estimate depend on the forms assumed for every rule.

Reported probabilities are coarse, and taking a reported zero literally imposes an absolute restriction, so the measurement model cannot be left implicit. The model used here follows the approach of inferring a respondent's rounding practice from their pattern of answers across questions and converting point reports into intervals (Manski and Molinari, 2010). When respondents to a Health and Retirement Study module on late-onset dementia were probed after their first answer, nearly half reported their probability as a range, which the authors interpret as an imprecise belief and not as a rounded precise one (Giustinelli et al., 2022). Focal-only respondents are therefore reported below as a group whose point-belief status is not identified. Once Dutch survival expectations are treated as partially identified, through the few points elicited and possible rounding, women's expectations can no longer be rejected as consistent with life tables, although point-identified survival curves place them below the tables (Bissonnette and de Bresser, 2018).

Whether decision-making quality is an individual characteristic has been studied for choice consistency, measured as a person-level ability that predicts outcomes outside the laboratory (Choi et al., 2014), and for forecasting accuracy that persists across years and questions in tournament settings (Mellers et al., 2015). Elicited behavioural biases are moderately stable within person over three years (Stango and Zinman, 2020) and correlated with one another within person (Stango and Zinman, 2023). The analogue for expectations, that rational-expectations consistency is a property of people and not of environments, needs repeated belief and realization pairs for the same person in several distinct domains. The available panels do not supply that structure, so the framework here bounds a population share without classifying individuals.

Finally, the magnitude is a functional of estimated cell means and weights that is Hadamard directionally but not fully differentiable, the setting for which Fang and Santos (2019) show that the standard bootstrap is consistent only at points of full differentiability and develop a valid alternative. Closely related procedures test whether a linear system with estimated components admits a solution, with the right-hand side estimated (Andrews et al., 2023; Fang et al., 2023) or with the coefficients estimated as well (Goff and Mbakop, 2025; Bai et al., 2026). These tests address existence. Here existence is handled through a route that is exact in finite samples under independence within cells, and magnitude through the numerical delta method of Hong and Li (2018), which builds on the framework of Fang and Santos (2019), implemented with multiplier draws and justified asymptotically. Design-based replication serves as a cross-check, and a simulation battery verifies the coverage of both.

3 Framework and Identification

3.1 Setup

Data. For each individual i the survey records a reported probability r_i that a binary event will occur within a stated horizon, and $Y_i \in \{0, 1\}$ indicates whether it does. In the application r_i is the percent chance, stated in 2004, of living to an individually assigned target age, and $Y_i = 1$ if the individual reaches it. The survey also records a vector W_i of characteristics, here the respondent's sex, age group, self-rated health and current smoking, which takes finitely many values. Two features of the data separate what is recorded from the objects the theory restricts. The first is that Y_i is not always resolved by the end of follow-up. Each individual is a known success ($R_{1i} = 1$), a known failure ($R_{0i} = 1$) or unresolved ($U_i = 1$), with $R_{1i} + R_{0i} + U_i = 1$, and 38.1 percent of the survival outcomes in the estimation sample are unresolved. The second is that r_i is recorded on a coarse grid, so it need not equal the individual's belief. The reporting rule of Section 5 maps each report into an interval $[a_i, b_i]$ that contains the belief.

The belief. Let $p_i \in [0, 1]$ denote the individual's subjective probability that $Y_i = 1$. Because the event is binary, this single number is the individual's whole subjective distribution over the outcome, a Bernoulli whose mean fixes its entire distribution function, so restricting means loses nothing that a median, a mode or a full distribution would add. The theory restricts p_i, and the data reveal it only up to the interval $[a_i, b_i]$.

What is not observed. The analyst does not observe the individual's information set I_i at the time of the report, how the individual formed the belief or how the individual came to know what they know. The framework makes no assumption about any of these. It does not model the acquisition of information, and nothing below depends on how I_i came about.

Where rationality enters. For a binary event, rational expectations in the sense of Muth (1961) implies $p_i = \mathbb{P}(Y_i = 1 \mid I_i)$, where $\mathbb{P}$ is the objective probability of the event given what the individual knows. This is the only economic restriction in the paper. Muth's hypothesis is broader, since it ties expectations to the predictions of the relevant economic theory and to the way information is used. The paper studies one observable implication of it and makes no assumption about preferences or learning.

The asserted information set. Let Z_i be a subvector of W_i that the analyst asserts the individual knew at the time of the report. An individual knows their own belief, so the maintained restriction is $\sigma(p_i, Z_i) \subseteq I_i$. The choice of Z belongs to the analyst, and individuals are not assumed to select it. It is an assumption about a lower bound on what the individual knows, and it requires only $Z \subseteq I$, never $Z = I$. By the law of iterated expectations, rational expectations then imply $\mathbb{E}[Y \mid p, Z] = p$. The restriction has content even when Z is empty. In that case it says $\mathbb{E}[Y \mid p] = p$, so among individuals who hold the same belief the event occurs at that rate. Each variable added

to Z imposes the same restriction within finer groups, so asserting more can only lower the bound, which is Theorem 1 below.

Not every variable qualifies. An admissible Z must be fixed before the report and known to the individual when they make it. Variables recorded later, or ones the individual may not know, such as an undiagnosed condition, are excluded. Three objects should be kept apart. The information set I_i is unobserved. The vector Z_i is what the analyst asserts the individual knew. The variables the individual actually used to form the belief are also unobserved, and knowing a variable does not imply using it. Rational expectations requires the belief to incorporate all relevant information in I_i, so a calibration failure within cells of Z means that the belief did not use Z correctly, while calibration relative to Z says nothing about the rest of I_i. The condition is therefore necessary for rational expectations and never sufficient.

The application uses five nested choices. The first, Z_0, is empty, and the others add in turn the respondent's own sex, age group, self-rated health and smoking, all of which a respondent knows about themselves at the time of the 2004 interview. These rungs are alternative assertions by the analyst about the same moment. They do not describe individuals acquiring information over time. Write $\mathcal{Z}$ for the finite set of values of Z and $X_i = (p_i, Z_i)$ for the conditioning cell.

The exact benchmark. Propositions 1 to 3 treat the benchmark in which the belief is recorded exactly, $r_i = p_i$, and every outcome is resolved. Beliefs then lie in a finite set $\mathcal{P} \subset [0,1]$, since reports are recorded on a grid, so (Y, p, Z) lives on the finite sample space $\{0,1\} \times \mathcal{P} \times \mathcal{Z}$, and every distribution P below is a probability distribution on that space. Cells with zero probability play no role, so conditional expectations given X are cell means. Propositions 4 and 5 drop both simplifications. The data then restrict the joint law of (Y, p, Z) only through the intervals $[a_i, b_i]$ and the resolved outcomes, and the bounds maximize over every complete-data law consistent with what is recorded.

Definition 1 (Calibration relative to an asserted information set)**.** Let $\mathcal{R}(Z)$ denote the set of distributions P on $\{0,1\} \times \mathcal{P} \times \mathcal{Z}$ such that

$$\mathbb{E}_P[Y \mid p, Z] = p \qquad P\text{-almost surely.} \tag{1}$$

A distribution P is calibrated relative to Z if $P \in \mathcal{R}(Z)$.

Condition (1) says that among individuals who hold the same belief and share the same value of Z, the event occurs at that rate. It is the sharp observable content of rational expectations under the assertion $\sigma(p, Z) \subseteq I$. It is necessary, by the law of iterated expectations. It is sufficient only in a weak sense. Any P satisfying (1) admits a rational-expectations representation, with information set $\sigma(p, Z)$, but the representation does not reveal what information an actual individual had or used. Throughout the paper, "rationalizable" means compatible with this observable implication under the stated Z, and the bound $\bar{\pi}$ defined below is the largest share of the population whose beliefs and outcomes admit such a representation. Whether any particular individual used their

information rationally is not identified.

Calibration is also distinct from accuracy. Accuracy measures how close a forecast comes to the realization under some loss function, and it rewards knowing more. Calibration asks only whether the event occurs at the stated rate among individuals who state that rate. A respondent who reports the base rate is calibrated even though a better-informed forecast would be more accurate, and two respondents who report the same probability and differ only in whether the event occurred have different realized accuracy and the same status under (1). The paper bounds calibration, which rational expectations implies, and makes no claim about accuracy.

Calibration on the belief alone is a weak restriction, since an individual who reports the unconditional base rate for every outcome satisfies it while knowing nothing. Enlarging Z asserts that individuals know more, and the resulting sequence of bounds is the object of interest here.

Let P_{obs} denote the population joint distribution of (Y, p, Z), a distribution on the same space that is observed directly in the exact benchmark. Suppose a fraction π of the population is calibrated in the sense of Definition 1 and the remainder behaves in a way the analyst is unwilling to restrict:

$$P_{\text{obs}} \;=\; \pi\, P_{\text{RE}} \;+\; (1-\pi)\, P_A, \qquad P_{\text{RE}} \in \mathcal{R}(Z), \qquad P_A \text{ unrestricted.} \tag{2}$$

Here P_{RE} is the distribution of (Y, p, Z) among calibrated individuals and P_A is any distribution on $\{0,1\} \times \mathcal{P} \times \mathcal{Z}$.

Proposition 1 (Identified set)**.** *Under* (2) *the identified set for* π *is the interval* $[\,0,\ \bar{\pi}(Z)\,]$*, where* $\bar{\pi}(Z) = \sup\{\pi : (2) \text{ admits a solution}\}$*. The lower bound equals zero for every* P_{obs}*.*

The lower bound is uninformative because P_A is unrestricted and may itself satisfy (1), so a world in which nobody belongs to the calibrated component and everybody merely looks calibrated is never refuted. Only the upper bound carries information. The estimand $\bar{\pi}$ is the *maximum rationalizable share* and $1 - \bar{\pi}$ is the *minimum non-rationalizable share.*

3.2 Identification

Fix a cell $x = (p, z)$ with $P_{\text{obs}}(X = x) > 0$ and let $m(x) = P_{\text{obs}}(Y = 1 \mid X = x)$ be the realized rate within it.

Proposition 2 (Sharp cell bound)**.** *The largest mass share within cell* x *that can satisfy* (1) *is*

$$\bar{\lambda}(p, m) \;=\; \min\left\{1,\ \frac{m}{p},\ \frac{1-m}{1-p}\right\}, \tag{3}$$

with a ratio whose denominator is zero read as $+\infty$*, and this value is attained.*

The logic is an accounting identity. Within the cell the rational component must have mean exactly p, while the contaminating component may have any mean in $[0,1]$, so $m = \lambda p + (1-\lambda)a$ for some $a \in [0,1]$. Maximizing λ subject to $a \in [0,1]$ delivers (3). The function is a tent in m, equal to one exactly when $m = p$ and falling away linearly on either side. It rises with slope $1/p$

below the peak and falls with slope $-1/(1-p)$ above it. Figure 1 plots it. The function is concave in m, which drives the monotonicity result, and its left arm is steep when p is small, so confident reports about rare events are the most informative.

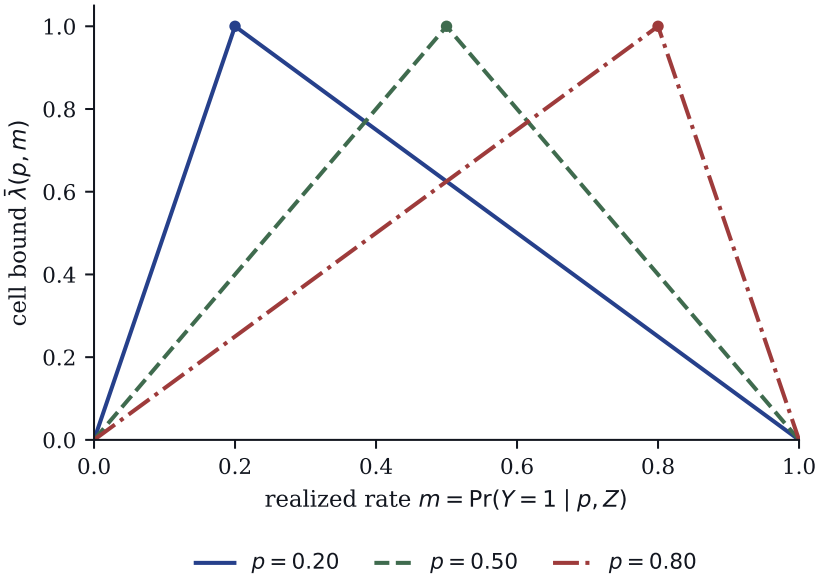


Figure 1: The sharp cell bound as a function of the realized rate

Notes. Each curve plots $\bar{\lambda}(p, m)$ from (3) against the realized rate m for one reported belief p, given in the legend. The bound equals one only where the realized rate coincides with the reported belief, shown by a dot. The steeper left arm for small p shows that low-probability reports place the tightest restrictions when the event nevertheless occurs.

Proposition 3 (Population bound)**.** *Let F_X denote the distribution of cells under P_{obs}, and for a cell $X = (p, z)$ write $p(X) = p$ for its reported belief. Then*

$$\bar{\pi}(Z) \;=\; \mathbb{E}_{F_X}\big[\,\bar{\lambda}\big(p(X),\, m(X)\big)\,\big]\,, \tag{4}$$

and the bound is sharp, in the sense that a pair (P_{RE}, P_A) attains it.

Sharpness follows because the constraints in (2) separate across cells. The rational component's conditional law given the cell is restricted only through its mean, so the cell-level maxima can be attained simultaneously, and the rational component's marginal over cells is then determined by the attained weights. The proof is in Appendix A.

Two features of survey data change the cell bound. Probabilities are reported on a coarse grid, so the belief is known only to lie in a respondent-specific interval $[a_i, b_i]$ inferred from the respondent's own answering pattern, and the bound maximizes over the beliefs that interval admits.

Proposition 4 (Interval beliefs)**.** *If the belief is known only to lie in* $[a, b]$*, the sharp cell bound is*

$$\bar{\lambda}(a, b, m) = \begin{cases} 1, & m \in [a, b], \\ m/a, & m < a, \\ (1-m)/(1-b), & m > b. \end{cases} \tag{5}$$

The tent of Proposition 2 becomes a plateau, with a consequence that recurs throughout the paper. On the plateau the bound is locally constant in m, hence locally differentiable with zero derivative, so the finite-sample bias of the plug-in estimator at a kink largely disappears. Interval beliefs add statistical regularity and reduce identifying power.

Some realizations are also unobserved. In the application a respondent who is still alive and has not yet reached the target age has an undetermined survival outcome. Within a cell, let m_1 denote the share of respondents known to have $Y = 1$ and u the share whose outcome is unresolved, so that the share $1 - m_1 - u$ is known to have $Y = 0$. Treating unresolved mass as freely allocable enlarges the set of realizations compatible with rationality.

Proposition 5 (Censoring)**.** *With belief interval* $[a, b]$*, known success mass* m_1 *and unresolved mass* u*,*

$$\bar{\lambda} \;=\; \begin{cases} 1, & [a, b] \cap [m_1,\, m_1 + u] \neq \emptyset, \\ (m_1 + u)/a, & a > m_1 + u, \\ (1 - m_1)/(1 - b), & b < m_1. \end{cases} \tag{6}$$

Theorem 1 (Information-set monotonicity)**.** *If* Z' *contains every variable in* Z*, written* $Z \subseteq Z'$*, then* $\bar{\pi}(Z') \leq \bar{\pi}(Z)$*.*

The argument is Jensen's inequality applied to a concave function. The bound $\bar{\lambda}(p, \cdot)$ is a minimum of two affine functions of m and therefore concave, and refining the conditioning set replaces $m(p, Z)$ by a collection of cell means whose average given (p, Z) is $m(p, Z)$. Hence $\mathbb{E}[\bar{\lambda}(p, m(p, Z')) \mid p, Z] \leq \bar{\lambda}(p, m(p, Z))$, and integrating gives the result. With interval beliefs and censoring the bound is concave in the cell's outcome shares and can only fall when the belief interval shrinks, and a refined cell's interval lies inside the interval of the cell it refines, so the same conclusion holds. Appendix A gives the argument. Theorem 1 makes the sequence $\bar{\pi}(Z_0) \geq \bar{\pi}(Z_1) \geq \cdots$ an object with content. The ordering holds in the population, so an observed non-monotonicity is an estimation problem.

The hypothesis that the entire population could be calibrated is a restriction cell by cell. Index the cells with positive probability by $c = 1, \ldots, C$, and write $[a_c, b_c]$ for the belief interval of cell c and $m_1(c)$ and $u(c)$ for its known-success and unresolved shares.

Proposition 6 (Full-RE feasibility)**.** $\bar{\pi}(Z) = 1$ *if and only if, for every cell* c*,*

$$m_1(c) \;\leq\; b_c \qquad \textit{and} \qquad m_1(c) + u(c) \;\geq\; a_c. \tag{7}$$

The first inequality says the cell cannot produce more successes than the most permissive admissible belief allows. The second says that even counting every unresolved case as a success, the cell cannot produce too few. Proposition 6 converts a statement about a population aggregate into a finite collection of statements about single binomial proportions, which makes the existence question exact.

4 Testing Existence versus Estimating Magnitude

The two questions this paper poses are not equally hard. Asking whether $\bar{\pi} < 1$ is an exact finite-sample problem, when observations within a cell are independent, that reduces to a finite collection of one-sided binomial hypotheses. Asking how far below one $\bar{\pi}$ lies is a partial-identification problem whose inference must cover every cell and every cell weight at once.

4.1 Existence

By Proposition 6, full rational expectations is feasible if and only if every cell satisfies (7). Rejecting it therefore requires only that *one* inequality fail. For each respondent let R_1 indicate that the outcome is known to have occurred, R_0 that it is known not to have occurred and $U = 1 - R_1 - R_0$ that it is unresolved, so that $P_{\text{obs}}(R_1 = 1 \mid c) = m_1(c)$ and $P_{\text{obs}}(U = 1 \mid c) = u(c)$. The two conditions in cell c are

$$P_{\text{obs}}(R_1 = 1 \mid c) \leq b_c \qquad \text{and} \qquad P_{\text{obs}}(R_0 = 1 \mid c) \leq 1 - a_c, \tag{8}$$

the second being equivalent to $P_{\text{obs}}(R_1 = 1 \mid c) + P_{\text{obs}}(U = 1 \mid c) \geq a_c$.

Each of (8) is a one-sided hypothesis about a single proportion, conditional on the cell count. Exact p-values follow from the binomial distribution without any asymptotic approximation:

$$p_c^{\text{high}} = \mathbb{P}[\text{Bin}(n_c, b_c) \geq k_{1c}]\,, \qquad p_c^{\text{low}} = \mathbb{P}[\text{Bin}(n_c, a_c) \leq k_{1c} + k_{uc}]\,. \tag{9}$$

Here n_c is the number of respondents in cell c, k_{1c} the number known to have reached the outcome and k_{uc} the number whose outcome is unresolved. The null is the intersection of all $2C$ inequalities, so the Holm step-down procedure applied to the collection controls the family-wise error rate in finite samples under arbitrary dependence across the tests. There is no linear program, no bootstrap, no delta method, no estimated cell weights and no simultaneous confidence region over primitives. The multiplicity burden is a step-down over $2C$ hypotheses, which is mild because a single decisive cell suffices.

Cells with fewer than twenty-five respondents are not tested, since a handful of observations cannot refute the inequality at any useful level. Restricting the family to a subset of cells leaves the global test valid, because rejecting on any tested cell still rejects the global null, and it can only cost power. The exactness of a component p-value is conditional on independent Bernoulli observations

within a cell, and household survey designs interview more than one member of a household, so the test is also reported below under two treatments that break that dependence.

4.2 Magnitude

The magnitude is an aggregate. When the cells partition the population, $w_c = P_{\text{obs}}(c)$ is the population share in cell c and $\bar{\lambda}_c$ is the sharp cell bound of Section 3 evaluated at cell c,

$$1 - \bar{\pi} \;=\; \sum_{c} w_c \left(1 - \bar{\lambda}_c\right), \tag{10}$$

with equality. The expression becomes an inequality only when inference replaces each term by a lower bound or restricts attention to a subset of cells.

Two routes to a one-sided lower bound on (10) are available. The first inverts the same exact binomials cell by cell and combines them with exact bounds on the weights. It is valid in finite samples with no asymptotics, but it is very conservative. The multiplicity correction runs over every cell proportion and every cell weight, and because each term is bounded at its own worst case the sum ignores that the cells cannot all be simultaneously extreme. In the application this route yields a bound roughly a fifth of the point estimate.

The second applies a design-based replication method to the aggregate directly. This recovers most of the lost precision, since resampling the design captures the joint behaviour of the cells. The map from the data to $\bar{\pi}$ involves minima, maxima and a binding constraint that can switch from one arm of (3) to the other, so the functional is directionally but not fully differentiable, and the standard bootstrap is consistent for such a functional only where it is fully differentiable (Fang and Santos, 2019). Validity must be assessed by simulation on known truths that preserve the survey design, the cell structure, the belief intervals and the censoring pattern. A third route removes the reliance on simulation for validity. Appendix B.4 develops a Gaussian multiplier form of the numerical delta method of Hong and Li (2018), built on the design's linearization covariance, states the conditions under which its one-sided bound is asymptotically valid, and supplies the certified lower bounds reported below, with replication retained as the design-based cross-check.

The asymmetry between the two problems follows from the structure of the question. Refuting a universal statement requires one counterexample, and a counterexample in a single cell is a statement about one binomial proportion. Quantifying how far a population is from the universal statement requires adding up evidence across all cells, and the addition must respect that all of them are estimated. A literature that rejects rational expectations routinely, and that rarely reports how much of a population those rejections implicate, is doing what the structure of the two problems predicts.

5 Data and Belief Measurement

5.1 Data

The application uses the Health and Retirement Study, a biennial panel of Americans over fifty that asks probabilistic expectations questions in a dedicated section of every core interview and follows respondents until death (Health and Retirement Study, 2025; RAND Center for the Study of Aging, 2025). The HRS is sponsored by the National Institute on Aging (grant numbers NIA U01AG009740 and NIA R01AG073289) and is conducted by the University of Michigan. Wave 7, fielded in 2004, is the anchor at which beliefs are elicited, and waves 8 through 16 supply realizations. Of 20,129 respondents interviewed at the anchor wave, 17,983 answer at least one probability question.

The belief is the respondent's own survival probability. Respondents are asked for the percent chance of living to a target age that the instrument adapts to their current age, so the question is individually tailored and not common across respondents. The adapted target is recorded, which makes the realization well defined at the individual level. The event is $Y_i = 1$ if respondent i reaches their own target age.

The realization is constructed from mortality and from observed ages in later waves. It resolves as one when the respondent is recorded as having died at or after the target age, or is observed alive at an interview having reached it. It resolves as zero when the respondent is recorded as having died before the target age. It is left *unresolved* when the respondent is alive at last observation but has not yet reached the target. Because the targets are set well beyond the respondent's age at interview, this group is large, at 38.1 percent of the estimation sample. Unresolved cases are never dropped. They enter as mass whose configuration may be allocated in whichever way is most favourable to rationality, following Proposition 5. Dropping them would select on the outcome being measured.

Estimation uses the complex survey design. Standard-error strata and half-sample codes give 56 strata with two half-samples each, and person-level analysis weights are used for all population quantities. Three restrictions define the weighted estimand. The weights represent community-dwelling older Americans in 2004, so institutionalized respondents, who carry a separate nursing-home weight, are outside it. Age-ineligible younger spouses carry zero weight, 817 respondents with a mean age of 47.7 against 65.5 among the weighted, leaving 12,745 of 13,562, or 94.0 percent, with a positive weight. And the estimation sample excludes respondents who report only on the focal grid, meaning that every probability they give is zero, fifty or one hundred, for the reason given below.

The estimand is therefore the community-dwelling older population of 2004 *among those who report probabilities on a grid finer than the focal one.* That last restriction is the largest. It removes a group for whom the object being bounded is not defined.

Table 1: Estimation sample, belief measurement and realization resolution

Panel A. Sample	
Respondents interviewed at the anchor wave	20,129
answering at least one probability question	17,983
in the authoritative estimation sample	13,562
with positive respondent weight	12,745
Households represented	9,939
Share in a household with two or more sample members (%)	53.0
Sampling-error strata	56
Finest-partition cells (atoms)	244
Panel B. Reporting granularity, of 17,983 with a classifiable grid	
Grid 1, some answer not a multiple of five	1,045 (5.8%)
Grid 5, multiples of five	5,318 (29.6%)
Grid 10, multiples of ten	7,600 (42.3%)
Grid 50, focal $\{0, 50, 100\}$ only	4,020 (22.4%)
Mean probability questions used per respondent	6.0
Survival-belief respondents meeting all other criteria	16,780
focal-only, excluded, status not identified	3,218 (19.2%)
Panel C. Survival realization	
Unresolved at last observation (%)	38.1
Weighted rate of reaching the target age, resolved cases	0.4180
Zero-weight respondents	817
mean age	47.7
mean age among positively weighted respondents	65.5

Notes. Health and Retirement Study, Wave 7 (2004), with realizations from Waves 8 through 16 (Health and Retirement Study, 2025; RAND Center for the Study of Aging, 2025). Panel B shares are of the 17,983 respondents with a classifiable reporting grid, which is the finest granularity a respondent displays across all subjective-probability questions answered at the anchor wave. The last two rows of Panel B use as denominator the respondents who report a survival belief and meet every other sample criterion, of whom the focal-only group is then excluded because their point-belief rational-expectations status is not identified. A realization is unresolved when the respondent is alive at last observation and has not yet reached their own target age. Unresolved cases are retained and their mass is allocated as in Proposition 5. Weighted quantities use the person-level respondent weight and describe the population that weight represents, among respondents whose reporting grid is finer than the focal one.

5.2 Reported probabilities

Subjective probabilities in household surveys are reported coarsely, and taking a reported zero literally imposes an absolute restriction, since a rational agent reporting zero must place no mass on the event and a single occurrence in such a cell is then unrationalizable. Since coarse reporting concentrates at zero, fifty and one hundred, the measurement model cannot be left implicit.

The rule is fixed in advance and follows the logic that a respondent's finest observed granularity reveals the grid on which they report. Using every subjective-probability question the respondent answered at the anchor wave, on average 6.0 of them, the reporting grid g_i is set to one if any answer is not a multiple of five, to five if any answer is a multiple of five but not of ten, to ten if any answer is a multiple of ten but not focal, and to fifty if every answer lies in $\{0, 50, 100\}$. A report x_i then becomes the interval $[x_i - g_i/2,\ x_i + g_i/2]$ truncated to the unit interval. A zero from a respondent who elsewhere reports thirty-seven and sixty-three stays essentially zero. A zero from a respondent who never leaves the focal grid becomes [0, 0.25]. Zeros and hundreds are not uniformly widened. Only 5.8 percent of respondents report at a resolution finer than multiples of five, and 22.4 percent never leave $\{0, 50, 100\}$.

The rule identifies how coarsely a respondent reports. It does not identify whether a focal-only respondent holds a precise point belief and rounds it, or holds a genuinely imprecise belief that no point summarizes. Those are different objects. The first is measurement error and is handled by the interval bound of Proposition 4. The second means that the benchmark $p = \mathbb{P}(Y = 1 \mid I)$ is not well defined for that respondent, so the point-belief version of rational expectations has no testable content for them. The authoritative specification therefore excludes focal-only respondents from point-belief inference and reports them separately as a group whose rational-expectations status is *not identified*, instead of counting them as rationalizable by default. Of the 16,780 respondents who report a survival belief and meet every other sample criterion, 3,218, or 19.2 percent, report only on the focal grid and are excluded on this ground. Specifications that retain them with wide intervals, and specifications that take all reports literally, are treated as robustness.

The cell bound of Section 3 is sharp for a cell defined by a single belief interval, but the estimation cells group respondents whose own intervals differ, and a cell is assigned the smallest interval containing its members' intervals. The maximization inside a cell may therefore select a belief that no individual member admits. That relaxation can only raise the rationalizable share, so the reported values of $\bar{\pi}$ are upper bounds on the sharp bound computed with respondent-specific intervals.

6 Main Results

6.1 Rejection of full rational expectations

Across 47 cells and 94 one-sided hypotheses, the exact test rejects the hypothesis that the whole population could be rational, and it rejects by a margin far beyond any conventional level (Table 2).

Retaining one respondent per household leaves the conclusion unchanged, and so does a household-clustered bootstrap that relaxes the within-cell independence the binomial requires, although the latter costs roughly twenty-six orders of magnitude in the adjusted p-value. A fourth route evaluates the same inequalities with standard errors from balanced repeated replication over the sampling-error strata and half-samples, so dependence is permitted at the level of the sampling-error half-samples, and the rejection stands there as well. A simulation under full rational expectations that preserves the observed cell sizes and censoring rates rejects at rate 0.000, so the test is conservative in this setting.

Table 2: Exact test of full rational expectations

Panel A. Inference routes

Route	Cells	Tests	Smallest Holm p	Decision at 1%
Exact binomial	47	94	2.5×10^{-50}	reject
One respondent per household	43	86	3.2×10^{-51}	reject
Household-clustered bootstrap	47	94	3.2×10^{-24}	reject
Half-sample replication	47	94	1.9×10^{-30}	reject

Panel B. Most violating cells

Belief interval	n	Reached / unresolved	Direction of violation
[0.97, 1.00]	203	55 / 83	too few possible survivors
[0.95, 1.00]	487	162 / 202	too few possible survivors
[0.00, 0.03]	322	40 / 106	too many survivors
[0.00, 0.05]	696	90 / 227	too many survivors

Notes. Each cell contributes the two one-sided exact binomial hypotheses of equation (8), and Holm's step-down procedure is applied to the full collection, which controls the family-wise error rate in finite samples under arbitrary dependence across tests. Cells with fewer than twenty-five respondents are not tested, which leaves the global test valid and can only cost power. Exactness of an individual p-value also requires independent observations within a cell, which the second and third rows of Panel A relax by retaining one respondent per household and by resampling households. The fourth row replaces the binomial with a one-sided normal approximation whose standard errors come from balanced repeated replication over the sampling-error strata and half-samples, so dependence is permitted at the level of the sampling-error half-samples. That route is design-aware and not exact, and the exact finite-sample statement belongs to the binomial row alone. A simulation under full rational expectations that preserves the observed cell sizes and censoring rates rejects at rate 0.000 at the ten, five and one percent levels. "Too many survivors" means the realized rate exceeds any belief the interval admits. "Too few possible survivors" means that even counting every unresolved case as a survivor the cell cannot reach the lowest admissible belief.

The violations are bidirectional and they sit where the restrictions bind, at beliefs far from one half. Figure 2 plots every cell's admissible belief interval against the range of realized rates consistent with its resolved and unresolved outcomes. Respondents whose reporting interval reaches no higher than three or five percent reach their target age at rates near thirteen percent. Respondents whose interval starts at ninety-five percent or above reach it at rates between twenty-seven and thirty-six percent, and even counting every unresolved case as a survivor they cannot approach the lowest belief their own interval admits. Those cells establish that the universal hypothesis fails. They carry no information about how much of the population is implicated.

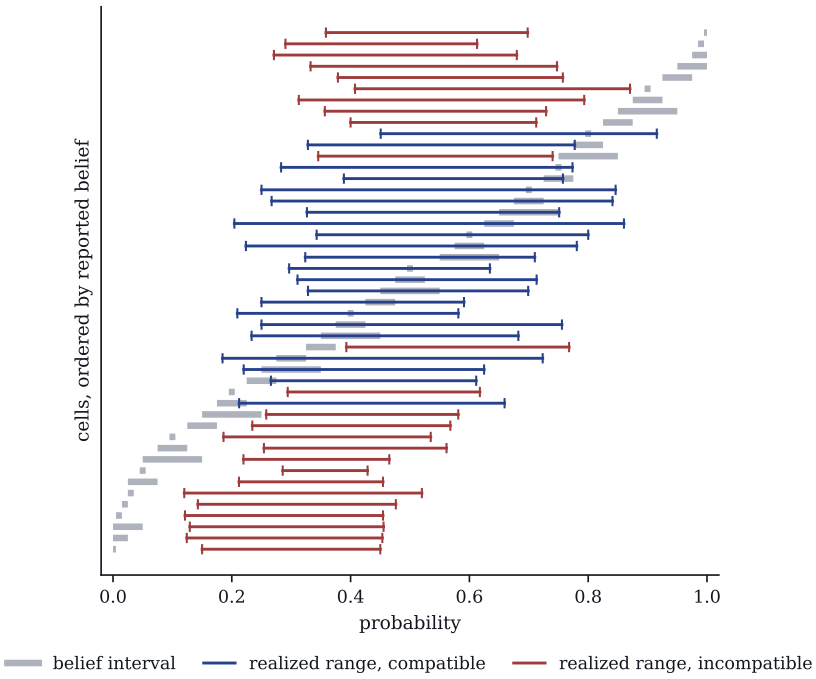


Figure 2: Admissible belief intervals against realized survival ranges, by cell

Notes. Each row is one cell of the authoritative specification, ordered by reported belief. The thick grey segment is the interval of beliefs the cell's respondents could hold given their reporting granularity. The thin segment is the range of realized rates compatible with the cell's resolved outcomes, running from the share known to have reached the target age to that share plus the share whose outcome is still undetermined. A cell is incompatible with rational expectations when the two segments do not overlap. Incompatible cells appear at both ends of the belief distribution. Cells with fewer than twenty-five respondents are omitted.

6.2 The information-set ladder

The ladder is a nested sequence of asserted information sets $Z_0 \subseteq Z_1 \subseteq \cdots \subseteq Z_4$, and each set is a rung. Z_0 is the reported belief alone, and each later rung adds one of the respondent's own sex, age group, self-rated health and current smoking, in that order. The cells of every rung are unions of a single finest partition of the sample into 244 atoms, where an atom is a group of respondents who share the same belief group, the reported probability rounded to the nearest tenth, and the same values of all four variables.

Conditioning on the reported belief alone, the maximum rationalizable share is $\bar{\pi} = 0.9776$, so the minimum non-rationalizable share is 2.24 percent of the weighted population, with a one-sided 95 percent lower bound of 1.82 percent from the multiplier procedure of Appendix B.4 and 1.86 from balanced repeated replication. Under the weakest assertion about what agents know, the data are consistent with roughly ninety-eight in every hundred people holding beliefs that some rational agent could hold. That number is a bound on composition.

It also conditions on a subpopulation, and the framework can drop that conditioning. The estimate above is computed among respondents whose point-belief status is defined, which excludes those who report only on the focal grid. Give the excluded group the most favourable treatment the framework allows, a cell bound of exactly one, and a bound for the whole eligible population follows, since the maximum rationalizable share over a union of groups is the weighted average of their bounds. Writing S for the weighted share whose status is identified,

$$1 - \bar{\pi}_{\text{all}} \;=\; S\left(1 - \bar{\pi}_{\text{identified}}\right). \tag{11}$$

The inequality runs the right way because their cell bound cannot exceed one. Here $S = 0.834$, so the minimum non-rationalizable share for the full eligible population is 1.87 percent at the base rung with a one-sided lower bound of 1.52, and 5.42 percent at the finest with a lower bound of 4.09. The inference treats the product directly, and replication gives 1.55 and 4.65 for the same two objects. The two objects answer different questions. The conditional one asks how much of the group whose reports admit a point belief cannot be rationalized. The population one asks how much of everyone.

Requiring beliefs to be efficient with respect to information the respondent unambiguously possesses moves the minimum non-rationalizable share from 2.24 percent to 6.50 percent, and the certified lower bound from 1.82 percent to 4.90 percent, so the floor nearly triples (Table 3 and Figure 3). The respondent's own sex moves nothing, from 0.9776 to 0.9776. Their age group produces the largest single increase, taking the non-rationalizable share from 2.24 to 4.78 percent. Self-rated health raises it to 5.84 percent and current smoking to 6.50 percent.

The bound does not move with the predictive content of a conditioning variable. Adding Z lowers $\bar{\pi}$ only when it uncovers heterogeneity in the compatibility between beliefs and realizations that aggregation over Z had concealed. A variable can predict mortality well and leave the bound untouched, provided beliefs already incorporate it correctly. So the reading of the ladder is that

age, self-rated health and smoking reveal conditional discrepancies between beliefs and outcomes that coarser cells mask, while sex reveals none at the point where the ladder adds it. The exercise measures incremental conditional rationalizability. Refining cells does not lower the bound automatically, as the sex rung shows, and it carries a finite-sample component, which the benchmark below measures.

Table 3: Maximum rationalizable share along the information-set ladder

		Weighted				Unweighted
Rung	Information set Z_j	$\bar{\pi}$	$1-\bar{\pi}$	95% LCB	BRR	$\bar{\pi}$
Z_0	belief only	0.9776	0.0224	0.0182	0.0186	0.9642
Z_1	+ own sex	0.9776	0.0224	0.0167	0.0188	0.9621
Z_2	+ own age group	0.9522	0.0478	0.0376	0.0424	0.9340
Z_3	+ own self-rated health	0.9416	0.0584	0.0454	0.0516	0.9227
Z_4	+ own current smoking	0.9350	0.0650	0.0490	0.0553	0.9147

Notes. $\bar{\pi}$ is the maximum share of the population whose beliefs can be rationalized as satisfying rational expectations with respect to the stated information set, and $1-\bar{\pi}$ is the corresponding minimum non-rationalizable share. Each rung conditions on the previous one plus the listed variable, all of which are attributes the respondent unambiguously knows about themselves. The ladder is nested by construction, since a single finest partition of 244 atoms is fixed in advance and every coarser rung is obtained by aggregating those same atoms, so the monotonicity of Theorem 1 holds exactly and not approximately. Weighted columns use the respondent weight and describe the population that weight represents among respondents whose reporting grid is finer than the focal one, which is the conditional estimand of Section 6. Lower confidence bounds are one-sided at 95 percent. The certified LCB column comes from the multiplier numerical delta method of Appendix B.4 with the step rule $\kappa_H = H^{1/4} = 2.74$, and the BRR column is the design-based cross-check from balanced repeated replication over the 56 sampling-error strata and their half-samples, using 64 replicates. The rungs are not causal. They are alternative assertions about what the analyst is willing to assume the agent knows.

Each rung is an alternative assertion about what the analyst is willing to assume agents know, and the sequence describes how much apparent rationalizability survives each assertion. Because monotonicity holds in the population, the informative quantity is the size of the decline. That reading requires a benchmark, because refining cells depresses a plug-in bound at a kink, and under literal beliefs that effect is large. The belief intervals of the authoritative specification turn the kink into the plateau of Proposition 4, which removes most of the bias. Simulating a population that is rational by construction, so that each respondent's own reported belief generates their outcome and the censoring pattern is held fixed, returns a benchmark of 0.00 percentage points at the first two rungs, 0.20 at the third, 0.55 at the fourth and 1.04 at the finest. Two quantities are distinct. The reported bound at Z_4 is 6.50 percent, and it is the object the inference of Table 4 covers. Its difference from the benchmark, about 5.46 points, says roughly a sixth of the reported figure is attributable to cell refinement. The base rung needs no such adjustment, since its benchmark is 0.00. Appendix F reports the corresponding benchmark under literal beliefs, where it is far larger and where a ladder could not be read directly.

Each increment along the ladder depends on where its variable enters, so the contribution of the four variables is also computed over all twenty-four orderings. Averaging the marginal

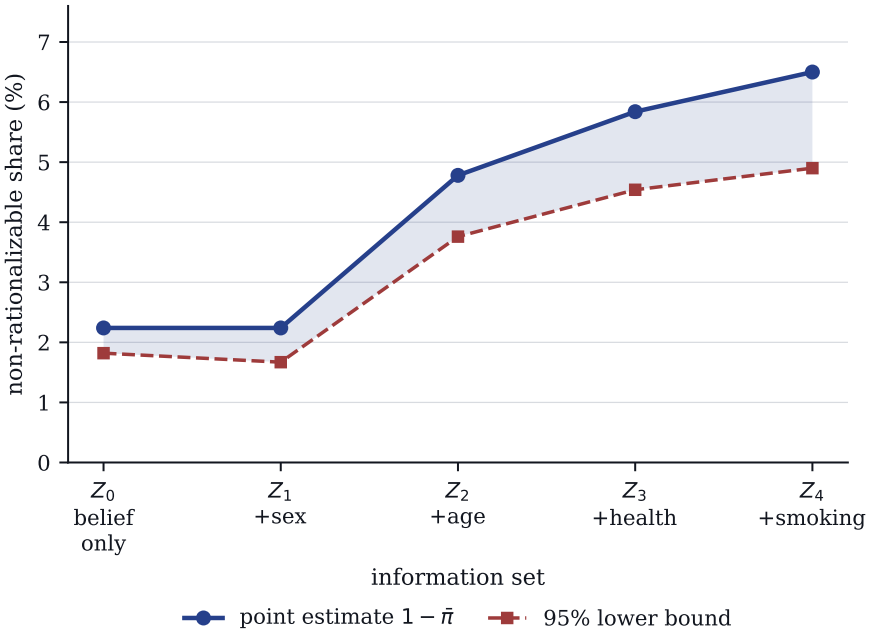


Figure 3: The minimum non-rationalizable share along the information-set ladder

Notes. Weighted estimates of $1-\bar{\pi}$ in percent, with the certified one-sided 95 percent lower confidence bounds from the multiplier numerical delta method of Appendix B.4. The replication bounds, which sit at or slightly above these, are in Table 3. Each rung conditions on the previous one plus the listed attribute. The ladder is nested by construction, so the decline in $\bar{\pi}$ follows from Theorem 1. A simulation in which every respondent is rational by construction, on the same cells and with the same censoring, returns a benchmark of 0.00 percentage points at the first two rungs rising to 1.04 at the finest, so a part of the movement shown here is a finite-sample consequence of refining cells.

contribution of a variable across orderings gives its Shapley share of the 4.25 percentage-point rise, with 2.66 points for age group, 0.77 for self-rated health, 0.52 for smoking and 0.30 for sex. Every contribution varies with position, so the question is by how much. Age dominates whatever the order, contributing between 2.31 and 3.13 points, a range that never approaches the other three. Sex is the only variable whose contribution can vanish. It is exactly zero when it enters first, as in Table 3, and reaches 0.74 points when it enters last. The null response to sex is therefore a property of the ladder as ordered, while the dominance of age survives every ordering.

The claims about which variable matters are claims about increments, so the increments carry their own inference and their own placebo. Formed inside each replicate, the rise on adding age group is 2.53 points with a one-sided lower bound of 2.01, on adding self-rated health 1.06 with a lower bound of 0.70, and on adding smoking 0.66 with a lower bound of 0.25. Sex adds 0.00.

The paper's standard is that a bound is reported as certified only where its coverage has been assessed for that object, and the increments are a different object from the levels. Applying the coverage designs of Table 4 to the increments gives a minimum coverage of 0.970 for the first increment and 0.967 for the second, against a nominal 0.95, and 0.927 and 0.810 for the third and fourth. Only the first two hold the level. So the data certify that conditioning on age alone reveals at least 2.01 additional percentage points of non-rationalizability. The health and smoking increments are reported as replication intervals.

Figure 4 sets each increment against what refinement alone would produce. Permuting the added variable within the cells of the previous rung preserves its number of categories, its marginal distribution and the resulting cell sizes, and destroys only its relation to beliefs and outcomes. Across 1,000 permutations the median increase is 0.07 points for age, 0.25 for health and 0.31 for smoking, so refinement alone moves the bound. The largest permutation draw reaches 0.20, 0.47 and 0.51 points, and no draw attains the observed increment for any of the three variables. Sex behaves exactly like a permuted variable, which is how a variable carrying no additional conditional discrepancy should behave.

In both rungs the cells that contribute at all cover about a third of the weighted population, 29.9 percent at the base rung and 33.0 percent at the finest. The two rungs implicate a similar share of the population, and they differ in how the shortfall is distributed among the cells that carry it. At the base rung only 4 of 11 cells contribute and the largest single cell carries 47.1 percent of the total, so removing the five largest eliminates it. At the finest rung 160 of 244 cells contribute, no single cell carries more than 3.8 percent, and removing the ten largest still leaves 4.70 of the 6.50 percentage points. The violation visible without conditioning is the work of a handful of cells at the tails of the belief distribution. What the information-set ladder adds is dispersed across many cells, and no small group of them accounts for it.

The conservative treatment of belief intervals raises one more question. Letting the interval enter the partition key, so that every cell is homogeneous in $[a, b]$ and the cell program can no longer pick a belief no member admits, is closer to the object Proposition 4 describes. It raises the estimate from 2.24 to 3.20 percent at the base rung and from 6.50 to 9.22 percent at the finest. That

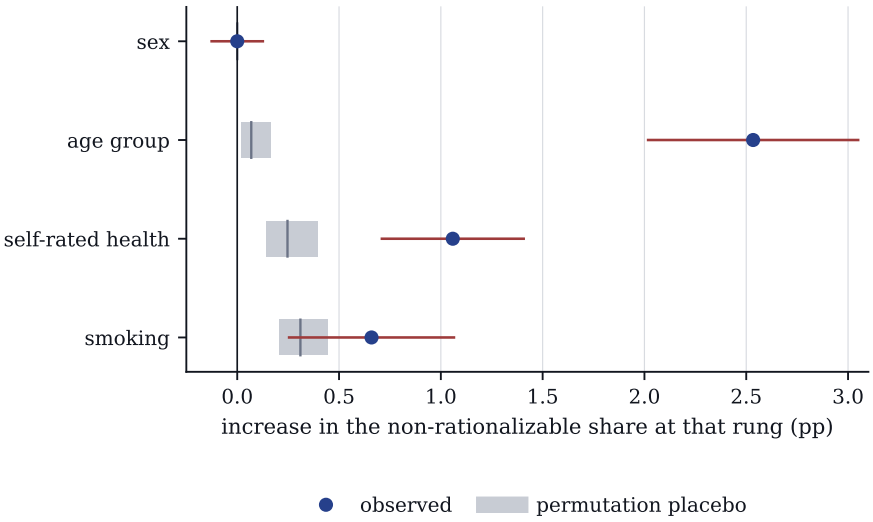


Figure 4: Observed information content against mechanical refinement

Notes. Points are the increase in the weighted minimum non-rationalizable share when the listed variable is added to the previous rung, in percentage points, with whiskers spanning the point estimate plus and minus 1.645 replication standard errors, so the lower end is the one-sided 95 percent bound. Grey bands run from the fifth to the ninety-fifth percentile of 1,000 permutations in which the variable is shuffled among respondents within the cells of the previous rung, which holds the number of categories, the marginal distribution and the cell sizes fixed while removing any relation to beliefs and outcomes. The tick inside each band is the permutation median. Increments are computed from unrounded levels.

estimate cannot be used. The sharp partition has 889 atoms with a median of 4 respondents, and on cells that small the plug-in bound is severely biased. Simulating a population that is rational by construction on the same cells returns a mean shortfall of 4.54 percentage points at the finest rung, so about half of the 9.22 is mechanical, and the one-sided lower bound covers in only 0.27 of replications where it should cover in 0.95. Sharpening the partition gains identification power and loses inferential reliability.

The two results together are the paper's organizing contrast. Universality is refuted decisively while the share the data require to be non-rationalizable is a few percent. They answer different questions.

The belief-measurement model moves the level of the bound and leaves the reading intact. Taking reports literally gives the tightest bound, since a reported zero or one hundred then imposes an absolute restriction, while restricting to respondents who report on a fine grid with no interval widening at all gives nearly the same answer as giving every respondent their own rounding interval. Two routes with opposite mechanics agree, which points to a real difference between precise and coarse reporters. Appendix F reports the three specifications.

Weighting moves the level of the bound and leaves its pattern unchanged. The unweighted ladder in the final column of Table 3 runs from 0.9642 to 0.9147, against 0.9776 to 0.9350 weighted, so weighting moves the bound toward less violation at every rung. The age-ineligible younger spouses

who carry zero weight contribute to that difference, and the ordering across rungs is identical in the two columns.

The design-based inference is assessed in two steps (Table 4). Balanced repeated replication agrees with the linearized variance to three decimals on four smooth statistics, which establishes that the replicate construction is sound. Coverage of the one-sided bound for the non-smooth object is then measured directly across simulation designs that span the boundary, interior shortfalls, a single binding constraint, the non-differentiable case, the empirical range and within-household outcome dependence up to a dependence parameter of 0.7. Against a nominal 0.95, the lowest coverage anywhere in the first-pass suite is 0.983, and in the stress grid calibrated to the empirical magnitude it is 0.990. These designs show no undercoverage where the functional is hardest. The procedure is conservative, with the mean bound running well below the truth in interior designs, so the reported figures understate the shortfall. The certified bounds of Table 3 come from the multiplier numerical delta method of Appendix B.4, which carries the asymptotic justification of Proposition 7 and whose minimum coverage in the same battery is 0.993. Replication gives slightly higher bounds at every rung, so the theory-backed route is also the marginally more conservative one.

Table 4: Validation of the design-based inference

Panel A. Replication against linearization, smooth statistics

Statistic	Estimate	BRR s.e.	Taylor s.e.
Weighted mean survival to target, resolved cases	0.4180	0.00761	0.00762
Weighted share in fair or poor health	0.2277	0.00517	0.00516
Weighted share reporting a belief of one half	0.1819	0.00481	0.00479
Weighted share aged eighty or above	0.0646	0.00274	0.00273

Panel B. Coverage of the nominal 95% *one-sided bound*

Simulation design	True $1 - \bar{\pi}$ at Z_4	Household ρ	Coverage
Boundary, every respondent rationalizable	0	0	1.000
Interior, shortfall spread across cells	0.0130	0	1.000
Single binding constraint	0.0102	0	1.000
Many cells at the non-differentiable point	0.0135	0	0.993
Empirical range	0.0500	0, 0.3, 0.7	≥ 0.991
Empirical value	0.0650	0, 0.3, 0.7	≥ 0.990
Above the empirical value	0.0800	0, 0.3, 0.7	≥ 0.996

Notes. Panel A compares balanced repeated replication over the 56 strata and their half-samples with the linearized variance for a stratified design with two units per stratum. Agreement on smooth statistics establishes that the replicate construction is sound but says nothing about a non-differentiable functional, which is why Panel B is necessary. Panel B simulates outcomes from designs whose true $1 - \bar{\pi}$ is known by construction while holding fixed the strata, half-samples, weights, the 244 atoms with their belief intervals and the censoring pattern. The last three rows use 1,000 Monte Carlo replications each and induce within-household outcome dependence through a shared-uniform copula that leaves every atom's marginal probability unchanged. The parameter ρ is the mixing probability of that copula, not a Pearson correlation. Coverage is reported at the finest rung. Against a nominal 0.95, the lowest coverage across all rungs is 0.983 in the first four designs and 0.990 in the last three. These designs show no undercoverage where the functional is hardest and do not establish uniform coverage over every data-generating process. The inference procedure was not modified in response to any of these results.

Survival is one domain. The same instrument asks respondents who have not yet reached sixty-two for the percent chance that they will be working full time after that age, and the same measurement architecture applies to it without modification. The outcome is full-time employment at the first interview at or after the target age, with death before it counted as not working, which leaves only 13.7 percent unresolved against 38.1 percent for survival. 4,078 respondents qualify and 3,476 carry a positive weight.

Full rational expectations is rejected in this domain too, over 28 cells and 56 one-sided hypotheses. The identified magnitude is far larger than for survival. Conditioning on the reported belief alone, the minimum non-rationalizable share is 11.54 percent with a one-sided 95 percent lower bound of 10.00 percent, against 2.24 percent for survival at the same rung. A rational population returns 0.01 percentage points on the same cells, so the estimate is not a small-sample artifact. Beliefs about working late in life are therefore harder to reconcile with what happens than beliefs about living.

Applying the base rung to every personal probability in the module whose realization the panel resolves cleanly puts that contrast in order. Figure 5 reports five domains. The two survival questions give the smallest shortfalls, 2.24 percent for the adapted target age and 1.78 for reaching seventy-five. Nursing-home entry within five years gives 4.17. The two employment questions give the largest, 8.32 percent for working after sixty-five and 11.54 for working after sixty-two. The rational-null benchmark does not exceed 0.01 percentage points in any of them, so the spread is not a cell-size effect.

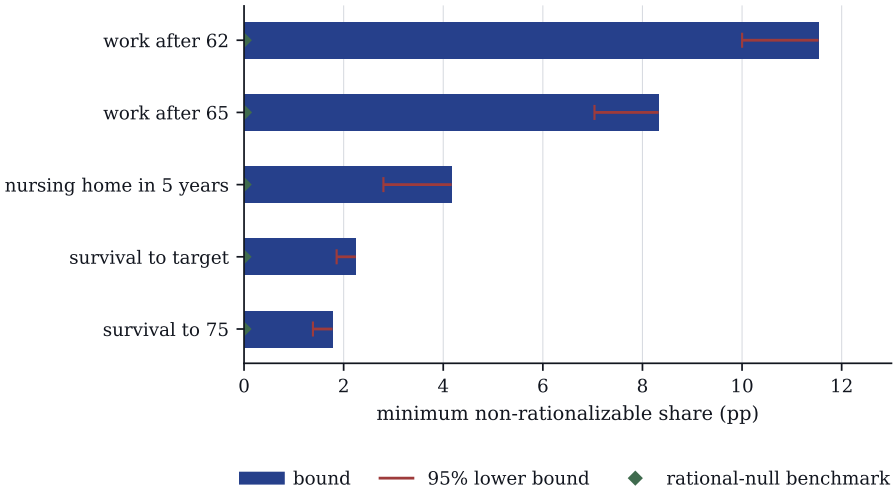


Figure 5: The minimum non-rationalizable share across expectation domains

Notes. Weighted estimates of $1 - \bar{\pi}$ conditioning on the reported belief alone, in percentage points, with the whisker running to the one-sided 95 percent lower bound from balanced repeated replication. Diamonds mark the shortfall a population that is rational by construction produces on the same cells. Each domain uses the measurement architecture of the survival application without modification and its own eligible sample. Bequest and inheritance items are excluded because the panel does not resolve them for surviving respondents.

Coverage of the lower bounds in Figure 5 at the boundary, where the truth is zero, is 1.000 in every domain. At an interior design calibrated to a shortfall of five percent, coverage is 0.990 and 0.980 for the two survival questions, 0.950 for nursing-home entry, and 0.900 and 0.920 for the two employment questions. The two employment bounds fall short of the nominal 0.95 at that design, so they are replication intervals and carry no certification.

It is a spread across five different samples, since each domain is asked of a different set of respondents over a different horizon. Holding the people fixed separates the domain from the sample. Estimating each domain and survival on exactly the same respondents gives the pairs in Table 5. Among the 3,397 respondents eligible for the work-after-sixty-two question, that domain gives 11.38 percent while survival gives 0.32 on the same people, so the gap is wider on a common sample than in the atlas. Among the 6,243 respondents eligible for the nursing-home question the ordering reverses, with nursing home at 4.56 percent and survival at 10.33.

The survival bound itself runs from 0.32 to 10.33 percent across these subsamples, against 2.24 in the full one. The maximum rationalizable share therefore depends on the domain and on the population, and the work-versus-survival contrast is partly a horizon effect. Among respondents under sixty-two, survival outcomes are mostly unresolved. The object is not a property of a population alone.

Table 5: Each domain and survival, estimated on the same respondents

Domain and its eligible sample	n	Domain	Survival
Survival to 75	6,781	1.80	0.68
Nursing-home entry in five years	6,243	4.56	10.33
Work full time after 65	4,077	8.33	0.38
Work full time after 62	3,397	11.38	0.32

Notes. Weighted minimum non-rationalizable share $1 - \bar{\pi}$ in percent, conditioning on the reported belief alone. Each row restricts to the respondents eligible for that domain who also report a survival belief on a non-focal grid, and estimates both bounds on exactly those people with the same architecture and the same weights. The survival column therefore varies across rows because the sample does. The rational-null benchmark is at or below 0.01 percentage points in every cell of the table.

The ladder is shorter in the work domain. That sample is under a third the size of the survival sample, the finest partition holds 232 atoms with a median of six respondents, and the rational-null benchmark rises from 0.01 percentage points at the base rung to 6.33 at the finest, against an estimate of 16.45. Only the base rungs are reported.

Eliciting two beliefs from the same respondents also supports a joint question, whether the same people can be rationalized in both domains at once. Appendix G develops the joint program, gives it a closed form for two domains, and shows that the joint restriction can cut a bound far below what the two marginal bounds imply. Applied to survival and work after sixty-two it adds 0.17 percentage points beyond the marginal envelope, so in these data the marginal bounds carry nearly all of the joint content.

7 Discussion

7.1 Interpretation

Proposition 1 fixes the interpretive rule. The identified set for the mixing weight runs from zero to $\bar{\pi}$, so the data never establish that anyone is rational. Every estimate in this paper is an upper bound on what could be, and its complement is a lower bound on what cannot be. This explains why the natural-sounding question, what fraction of people are rational, has no answer in these data, and it rules out reading successive rungs of the ladder as progressively identifying irrational individuals. Each rung shrinks the set of populations consistent with the data.

Scoring individuals by the accuracy of their probabilistic forecasts would not recover the missing information. Two respondents who both report a ten percent chance and who differ only in whether the event occurred have different realized accuracy and identical rationality. A score built from realized outcomes confounds whether the belief was rational, how much private information the individual had, and the irreducible randomness of the outcome. Only the first is the object of interest, and with one realization per person the three are not separable. Individual latent type membership is therefore not identified at any sample size.

7.2 Scope

Calibration on the belief alone is already a testable restriction, and it is a weak one, because reporting the unconditional base rate always satisfies it. Each assertion about what people know strengthens it, and the assertions are the analyst's. The ladder makes them visible and shows how much they move the answer. Requiring efficiency with respect to a respondent's own sex, age, self-rated health and smoking status nearly triples the share the data force to be non-rationalizable. The rungs used here were chosen because they are difficult to dispute. Nobody argues that a respondent does not know their own age or whether they smoke. A researcher willing to assert more, that agents know their own diagnosed conditions or their parents' longevity, would obtain a weakly lower bound by Theorem 1, and one willing to assert less would obtain a weakly higher one. Reporting the sequence presents an object defined only relative to such a choice.

The weighted estimand is narrower than the population the survey weights represent. It is older Americans living in the community, excluding age-ineligible younger spouses, who carry zero weight and contribute to the difference between the weighted and unweighted columns of Table 3, and excluding institutionalized respondents, for whom the study supplies a separate weight. It also excludes the 19.2 percent of otherwise eligible respondents who report only on the focal grid. That group is removed because the point-belief version of the hypothesis has no content for them, and no reweighting can restore them, because they are defined by the absence of the object being bounded. Equation (11) recovers a statement about everyone by giving that group the most favourable treatment the framework permits, giving 5.42 percent against 6.50 at the finest rung. Survival is one of the few domains in which respondents hold private information the analyst cannot observe, and it is absorbing, which makes the censoring treatment of Proposition 5 necessary. Expectations about

working after sixty-two behave differently on both counts, resolve for almost every respondent, and produce a shortfall five times the survival one at the same rung, so survival is the conservative domain.

8 Conclusion

Rejecting rational expectations and determining how much of a population cannot hold rational expectations are different questions with different answers. The first reduces to a finite collection of one-sided binomial inequalities, one pair per cell, and a single decisive cell settles it. In the Health and Retirement Study it is settled by a wide margin. Respondents who assign themselves almost no chance of reaching their own target age reach it far too often, and respondents who assign themselves near certainty reach it far too rarely, and neither pattern can be reconciled with any belief their reporting granularity admits.

The second question requires aggregating partial information across every cell and every cell weight, and the answer is correspondingly less precise and much smaller. Among older Americans in 2004 who report probabilities on a grid finer than the focal one, at least 1.82 percent cannot be rationalized as holding rational expectations about their own survival when nothing is assumed about what they know beyond their own reported belief. Requiring that beliefs also be efficient with respect to their own sex, age, self-rated health and smoking raises that floor to 4.90 percent, or about five people in a hundred. Treating the respondents whose reports admit no point belief in the way most favourable to rationality carries the same statements to the whole eligible population at 1.52 and 4.09 percent. How many people could have rational expectations depends on three things at once, and the framework makes each of them an object of estimation instead of an implicit modelling choice. It depends on the information rationality is required to use, which the ladder measures. It depends on what is being forecast, since on the same respondents beliefs about working after sixty-two leave 11.38 percent unrationalizable where survival beliefs leave 0.32. And it depends on the population, since the survival bound itself ranges from 0.32 to 10.33 percent across the subsamples that different questions define. There is no single rationalizable fraction to report.

The analysis delivers a bound on the composition of the population, sharp as an identification result, whose lower end is always zero. A decisive rejection of universal rationality and a shortfall of a few percent are answers to two questions that the literature has treated as one.

What the bound leaves open matters beyond this application. Rationalizability says which models the observed distribution can be reconciled with, and not how many agents are of a given type, so the identified set for that share runs from zero to the bound however large the sample grows. Any question that depends on how people would revise beliefs in an environment the data do not contain therefore needs more than compatibility with rational expectations in the environment they do. The framework makes the size of that gap measurable, which one needs before assuming it away.

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A Proofs

Throughout, $x = (p, z)$ indexes a cell, F_X is the distribution of cells, and $m(x) = P_{\text{obs}}(Y = 1 \mid X = x)$.

Proof of Proposition 1

Take $\pi = 0$ and $P_A = P_{\text{obs}}$. Condition (2) holds trivially and P_A is unrestricted, so 0 belongs to the identified set for every P_{obs}. The set of feasible π is an interval. If π' is feasible with decomposition (P_{RE}, P_A) and $\pi < \pi'$, then π is feasible with the same P_{RE} and with alternative component $\big[(\pi' - \pi)P_{\text{RE}} + (1 - \pi')P_A\big]/(1 - \pi)$, which is a probability measure because $\pi < \pi' \leq 1$. The identified set is therefore $[0, \bar{\pi}]$ with $\bar{\pi}$ the supremum of feasible values. □

Proof of Proposition 2

Within cell x, let λ denote the mass share of the rational component and let $a \in [0, 1]$ denote the mean of Y under the alternative component. The rational component must satisfy $\mathbb{E}[Y \mid x] = p$ by Definition 1, so

$$m \;=\; \lambda p + (1 - \lambda)a.$$

Solving for a gives $a = (m - \lambda p)/(1 - \lambda)$ when $\lambda < 1$. The constraint $a \geq 0$ is equivalent to $\lambda p \leq m$, and $a \leq 1$ is equivalent to $\lambda(1 - p) \leq 1 - m$. Hence $\lambda \leq m/p$ and $\lambda \leq (1 - m)/(1 - p)$, and with $\lambda \leq 1$ this yields $\lambda \leq \bar{\lambda}(p, m)$ as in (3).

Attainment is by construction. Set $\lambda = \bar{\lambda}(p, m)$. If $m \geq p$ take $a = 1$, which satisfies the identity because $\bar{\lambda} = (1 - m)/(1 - p)$ in that case. If $m < p$ take $a = 0$, which satisfies it because $\bar{\lambda} = m/p$. If $m = p$ take $\lambda = 1$. The rational component's conditional law given the cell is any distribution on $\{0, 1\}$ with mean p, which exists. □

Proof of Proposition 3

The restriction (1) is a conditional-mean restriction given X, so it binds cell by cell and imposes no cross-cell constraint. Let $\lambda(x)$ be the rational mass share in cell x. Feasibility of $(\lambda(x))_x$ is equivalent to $\lambda(x) \leq \bar{\lambda}(p(x), m(x))$ for F_X-almost every x, by Proposition 2. The total rational share is $\int \lambda(x)\, dF_X(x)$, which is maximized pointwise, giving (4).

For sharpness, set $\lambda^*(x) = \bar{\lambda}(p(x), m(x))$ and $\bar{\pi} = \int \lambda^* dF_X$. Define P_{RE} by giving cell x the mass $\lambda^*(x)\, dF_X(x)/\bar{\pi}$ and the conditional law on Y constructed in the proof of Proposition 2. Define P_A as the residual, which is a non-negative measure of total mass $1 - \bar{\pi}$ by construction and hence a probability measure after normalization. Then (2) holds with $\pi = \bar{\pi}$ and $P_{\text{RE}} \in \mathcal{R}(Z)$. □

Proof of Proposition 4

With the belief known only to lie in $[a, b]$, the cell bound is $\max_{p \in [a,b]} \bar{\lambda}(p, m)$. As a function of p the map m/p is decreasing and $(1 - m)/(1 - p)$ is increasing, so their minimum is single-peaked with

maximum at $p = m$, where it equals one. If $m \in [a, b]$ the unconstrained maximizer is admissible and the value is one. If $m < a$ the objective is decreasing on $[a, b]$, so the maximum is at $p = a$, where $m/a < 1 \leq (1 - m)/(1 - a)$ and the value is m/a. If $m > b$ the objective is increasing on $[a, b]$, so the maximum is at $p = b$, where $(1 - m)/(1 - b) < 1$ and the value is $(1 - m)/(1 - b)$. □

Proof of Proposition 5

Let m_1 be the known success mass, $m_0 = 1 - m_1 - u$ the known failure mass and u the unresolved mass, which may be split as $u = u_0 + u_1$ with $u_j \geq 0$. Given a belief p, a rational mass S requires success mass Sp drawn from $m_1 + u_1$ and failure mass $S(1 - p)$ drawn from $m_0 + u_0$, so

$$Sp \leq m_1 + u_1, \qquad S(1 - p) \leq m_0 + u_0, \qquad u_0 + u_1 = u.$$

Adding the two inequalities gives $S \leq m_1 + m_0 + u = 1$. If the two can be made to hold with equality simultaneously then $S = 1$, which requires $u_1 = p - m_1 \in [0, u]$, that is $m_1 \leq p \leq m_1 + u$. If $p > m_1 + u$ the binding constraint is the first with $u_1 = u$, giving $S \leq (m_1 + u)/p$. If $p < m_1$ the binding constraint is the second with $u_0 = u$, giving $S \leq (m_0 + u)/(1 - p) = (1 - m_1)/(1 - p)$. Maximizing over $p \in [a, b]$ as in the proof of Proposition 4 yields (6). □

Proof of Theorem 1

Fix p and let $Z \subseteq Z'$. The map $m \mapsto \bar{\lambda}(p, m)$ is the minimum of two affine functions and of the constant one, hence concave on $[0, 1]$. By the tower property $\mathbb{E}[\, m(p, Z') \mid p, Z \,] = m(p, Z)$, so Jensen's inequality for concave functions gives

$$\mathbb{E}\big[\bar{\lambda}\big(p, m(p, Z')\big) \,\big|\, p, Z\big] \;\leq\; \bar{\lambda}\big(p, \mathbb{E}[m(p, Z') \mid p, Z]\big) \;=\; \bar{\lambda}\big(p, m(p, Z)\big).$$

Taking expectations over (p, Z) and applying Proposition 3 at each information set gives $\bar{\pi}(Z') \leq \bar{\pi}(Z)$.

With interval beliefs and censoring, a cell carries an interval $[a, b]$ and outcome shares (m_1, u), and by Proposition 5 its bound is

$$\bar{\lambda}(a, b, m_1, u) = \min\left\{1,\ \frac{m_1 + u}{a},\ \frac{1 - m_1}{1 - b}\right\},$$

with a ratio whose denominator is zero read as $+\infty$. This expression coincides with (6) in each of its three cases. Two properties of this function deliver the result. For fixed $[a, b]$ it is the minimum of a constant and two affine functions of (m_1, u), hence jointly concave in (m_1, u). And it is non-increasing in a and non-decreasing in b, so shrinking the interval can only lower it. When a cell c is refined into cells c_k with population shares w_k, the outcome shares average back to the parent, $\sum_k w_k \, (m_1(c_k), u(c_k)) = (m_1(c), u(c))$ with $\sum_k w_k = 1$. In the application each cell is assigned the

smallest interval containing the intervals of its members, so $[a_{c_k}, b_{c_k}] \subseteq [a_c, b_c]$ for every k. Hence

$$\sum_k w_k \, \bar{\lambda}\big(a_{c_k}, b_{c_k}, m_1(c_k), u(c_k)\big) \;\le\; \sum_k w_k \, \bar{\lambda}\big(a_c, b_c, m_1(c_k), u(c_k)\big) \;\le\; \bar{\lambda}\big(a_c, b_c, m_1(c), u(c)\big),$$

where the first inequality uses the monotonicity in the interval and the second uses concavity and Jensen's inequality. Summing over parent cells gives $\bar{\pi}(Z') \le \bar{\pi}(Z)$. □

Proof of Proposition 6

By Proposition 3, $\bar{\pi} = 1$ if and only if $\bar{\lambda} = 1$ for F_X-almost every cell. By Proposition 5, $\bar{\lambda} = 1$ in a cell if and only if $[a_c, b_c] \cap [m_1(c), m_1(c) + u(c)] \neq \emptyset$, which holds if and only if $m_1(c) \le b_c$ and $m_1(c) + u(c) \ge a_c$. The second is equivalent to $\mathbb{P}(R_0 = 1 \mid c) = 1 - m_1(c) - u(c) \le 1 - a_c$. □

Proofs for the joint-domain program are in Appendix G.

B Inference

B.1 The exact test

By Proposition 6, $\bar{\pi} = 1$ is equivalent to the conjunction over cells of $m_1(c) \le b_c$ and $m_1(c) + u(c) \ge a_c$. Conditional on the cell count n_c, the number of resolved successes k_{1c} is binomial under the null boundary $m_1 = b_c$, and the number of cases compatible with success, $k_{1c} + k_{uc}$, is binomial under the null boundary a_c. The exact one-sided p-values are those in (9). Every tested cell contributes both hypotheses, so the family has exactly $2C$ members. When $b_c = 1$ the upper inequality is vacuous and when $a_c = 0$ the lower one is, and in those cases the component p-value is set to one and the hypothesis stays in the family. Retaining it leaves the family size fixed and can only make the step-down more conservative. Cells with fewer than twenty-five respondents are excluded from the family, the same threshold used for Figure 2. Testing a subset of the cells preserves validity of the global test and reduces power only.

The global null is the intersection of the $2C$ component nulls. Holm's step-down procedure applied to the ordered p-values controls the family-wise error rate at the nominal level in finite samples under arbitrary dependence across the tests (Holm, 1979). Rejection of any component rejects the global null.

Exactness of a component p-value requires independent Bernoulli observations within the cell, which household surveys violate. Appendix F reports the test restricted to one respondent per household, selected by a deterministic identifier rule that is blind to the outcome, and under a bootstrap that resamples households. A design-aware route in Table 2 evaluates the same inequalities with balanced-repeated-replication standard errors over the strata and half-samples, and it is a normal approximation. A simulation under full rational expectations that preserves the observed cell sizes and censoring rates, with the belief placed at the low endpoint, the midpoint and the high endpoint of each interval, rejects at rate 0.000 at the ten, five and one percent levels, so the procedure is conservative in this configuration.

B.2 Inference on the magnitude

The magnitude satisfies (10) with equality when the cells partition the sample and the $\bar{\lambda}_c$ are sharp. A lower confidence bound therefore requires simultaneous control over every cell proportion and every cell weight. Two routes were implemented.

The exact route places a Clopper–Pearson interval on each cell proportion and each cell weight at level α/K with K the number of intervals (Clopper and Pearson, 1934), and evaluates the bound at the least favourable point of the resulting box. Validity is immediate, since with probability at least $1-\alpha$ the truth lies in the box and on that event the true magnitude is at least the minimized value. It is severely conservative, because bounding each term at its own worst case ignores that the cells cannot be simultaneously extreme, and it is exact under independent sampling for the unweighted empirical distribution, so its target is the unweighted sample. The multiplier procedure of Appendix B.4 is the inferential route for the weighted target and supplies the certified bounds reported in the main text, with balanced repeated replication as the design-based cross-check.

The design-based route applies balanced repeated replication to the aggregate. This is sharper. The map from cell masses to $\bar{\pi}$ involves minima and a constraint that can switch arms, so the functional is directionally but not fully differentiable and standard resampling need not be consistent for it (Fang and Santos, 2019). Agreement between replication and linearization on smooth statistics, reported in Table 4, is necessary, and coverage is assessed directly by simulation.

B.3 Balanced repeated replication

The HRS supplies standard-error strata and, within each, two half-samples designed for replication or for an approximate two-per-stratum linearization. With $H = 56$ strata, a Sylvester Hadamard matrix of order $R = 64$ assigns, in replicate r and stratum h, one half-sample the weight $2w_i$ and the other zero. The variance estimator is

$$\widehat{\mathrm{Var}}(\hat{\theta}) = \frac{1}{R}\sum_{r=1}^{R}\left(\hat{\theta}_r - \hat{\theta}\right)^2, \tag{12}$$

and the one-sided lower bound is $\hat{\theta} - z_{1-\alpha}\,\widehat{\mathrm{se}}$ (Wolter, 2007).

For a stratified design with two units per stratum the linearized variance collapses to $\sum_h (u_{h1} - u_{h2})^2$, where u_{hi} is the unit total of the linearized variable. The replication and linearized estimators agree to three decimal places on all four smooth statistics in Panel A of Table 4.

B.4 A multiplier numerical delta method

The replication route is validated by simulation. This subsection develops a companion procedure whose justification is asymptotic, so that the simulations verify performance. It is a multiplier form of the numerical delta method of Hong and Li (2018).

Fix the finest partition and let θ collect, for every atom, the weighted mass, resolved-success mass and unresolved mass, each expressed as a share of the total analysis weight, so that θ is a

vector of order one whose sampling error vanishes at the rate of the design. The belief intervals are fixed by the partition, so the magnitude at each rung is a known map $\psi_j(\theta) = 1 - \bar{\pi}(Z_j)$. The map is invariant to a common rescaling of θ, so the implementation may equivalently form the unnormalized weighted totals and their contrasts, which changes nothing to first order.

Lemma 1 (Directional differentiability)**.** *On the set where every cell total is positive, each ψ_j is Lipschitz continuous and Hadamard directionally differentiable.*

Proof. The censored cell bound of Proposition 5 can be written $\bar{\lambda} = \min\{1,\ (m_1 + u)/a,\ (1 - m_1)/(1 - b)\}$, where an arm whose denominator is zero, $a = 0$ for the second or $b = 1$ for the third, is never binding in (6) and is simply omitted from the minimum, so each cell's bound is the minimum of the remaining finitely many terms, all with strictly positive denominators fixed by the partition. The cell shares (m_1, u) are continuously differentiable functions of the cell totals where the totals are positive, and the cell totals are linear in θ. A pointwise minimum of finitely many continuously differentiable functions is locally Lipschitz and Hadamard directionally differentiable, both properties are preserved by composition with continuously differentiable maps and by the chain rule for Hadamard directional derivatives (Fang and Santos, 2019), and ψ_j is a weighted average of such cell bounds whose weights are smooth in θ on the same set. □

Assumption 1 (Design asymptotics)**.** The atom partition is fixed, so the dimension of θ does not grow along the asymptotic sequence, and every cell total entering ψ_j is bounded away from zero. There is a rate $r_n \to \infty$ with $r_n(\hat{\theta} - \theta) \to_d N(0, \Sigma)$, and the linearization estimator $\widehat{\Sigma} = \sum_h d_h d_h'$, with d_h the half-sample contrast of the totals in stratum h, satisfies $r_n^2\,\widehat{\Sigma} \to_p \Sigma$.

Assumption 1 is the standard first-order theory for stratified multistage designs with two half-samples per stratum, in which the number of strata grows (Krewski and Rao, 1981; Wolter, 2007). It concerns only the linear statistics $\hat{\theta}$, where nothing non-smooth has yet happened.

The procedure draws Gaussian multipliers instead of treating the finite set of replicates as a bootstrap distribution. With $g_{s1}, \ldots, g_{sH}$ independent standard normals,

$$Z_s^* \;=\; \sum_h g_{sh}\, d_h, \qquad D_s \;=\; \frac{\psi\big(\hat{\theta} + \kappa_n Z_s^*\big) - \psi(\hat{\theta})}{\kappa_n}, \tag{13}$$

so that conditional on the data $Z_s^* \sim N(0, \widehat{\Sigma})$, and the one-sided bound is

$$\mathrm{LCB} \;=\; \psi(\hat{\theta}) \;-\; q_{1-\alpha}(D), \tag{14}$$

where $q_{1-\alpha}(D)$ is the empirical $(1-\alpha)$ quantile of the draws D_s themselves, not of $r_n D_s$. The rate never needs to be computed, because the unscaled draws Z_s^* already carry the sampling scale of $\hat{\theta}$.

Proposition 7 (Asymptotic validity)**.** *Let Lemma 1 apply at θ and let Assumption 1 hold. If $\kappa_n \to \infty$ and $\kappa_n / r_n \to 0$, then, with $q_{1-\alpha}(D)$ as in (14), $\mathbb{P}\big[\,\psi_j(\theta) \geq \psi_j(\hat{\theta}) - q_{1-\alpha}(D)\,\big] \to 1 - \alpha$ at every θ at which the law of the directional derivative $\psi'_{j,\theta}(G)$, $G \sim N(0, \Sigma)$, is continuous and strictly increasing at its $(1-\alpha)$ quantile.*

Proof. Write $\hat{h}_s = r_n Z_s^*$ and $\varepsilon_n = \kappa_n / r_n$, so that $D_s = r_n^{-1}\big[\psi(\hat{\theta} + \varepsilon_n \hat{h}_s) - \psi(\hat{\theta})\big]/\varepsilon_n$. Conditional on the data $\hat{h}_s \sim N(0, r_n^2 \widehat{\Sigma})$, which converges in probability to the law of G by Assumption 1, so the draws satisfy the bootstrap-consistency requirement for the underlying vector. Since ψ_j is Lipschitz and Hadamard directionally differentiable and $\varepsilon_n \to 0$ with $r_n \varepsilon_n = \kappa_n \to \infty$, the numerical delta method of Hong and Li (2018, Theorem 3.1), which builds on Fang and Santos (2019), gives that the conditional law of $\big[\psi(\hat{\theta} + \varepsilon_n \hat{h}_s) - \psi(\hat{\theta})\big]/\varepsilon_n$ converges in probability to the law of $\psi'_{j,\theta}(G)$. The empirical quantile of $r_n D_s$ therefore converges to $q_{1-\alpha}\big(\psi'_{j,\theta}(G)\big)$ wherever that quantile is a continuity point, and the coverage statement follows from $r_n\big(\psi_j(\hat{\theta}) - \psi_j(\theta)\big) \to_d \psi'_{j,\theta}(G)$, which is the delta method for directionally differentiable maps. □

The coverage statement is pointwise in θ. The finite-sample choice of the step is assessed with the coverage battery used for replication, whose boundary and kink designs probe the configurations where a plain bootstrap fails. The step is a rule. With asymptotics in the number of strata and r_n of order $\sqrt{H}$, the rule $\kappa_H = H^{1/4}$ satisfies both rate conditions, since $\kappa_H \to \infty$ and $\kappa_H / \sqrt{H} = H^{-1/4} \to 0$. In the implementation $H = 56$, so $\kappa_H = 2.74$, the draws number 4,000, and fixed steps $\kappa \in \{1, 2, 4\}$ are reported as sensitivity, where $\kappa = 1$ is the plain multiplier bootstrap.

The population bound of equation (11) is covered by the same theory. Extending θ with the weighted focal-only total makes the identified share S a smooth ratio of totals, the population magnitude is that ratio times ψ_j, and Lemma 1 and Proposition 7 apply to the product unchanged.

Applied to the estimation sample at $\kappa_H = 2.74$, the multiplier procedure reproduces the point estimates of Table 3 exactly and delivers the certified lower bounds reported in the main text, 1.82 percent at the base rung against 1.86 from replication, and 4.90 percent at the finest rung against 5.53, with 1.52 and 4.09 for the population bound against 1.55 and 4.65. The multiplier bounds sit at or slightly below the replication bounds at every rung. In the same coverage battery used for replication, the minimum coverage over every design and all five rungs is 0.993 at the rule step and 0.997 for the population bound, against a nominal 0.95. The battery simulates outcomes while holding the weights, the design and the focal-only mass fixed. The plain multiplier $\kappa = 1$ covers at 0.987 with bounds within two tenths of a percentage point of replication, and moving the step across $\{1, 2, 4\}$ shifts the finest-rung bound by about half a percentage point in total.

B.5 Coverage validation and stress tests

Coverage is measured by simulating outcomes from designs whose true $1 - \bar{\pi}$ is known by construction while holding fixed the strata, half-samples, weights, the 244 atoms with their belief intervals, and the censoring pattern. Only outcomes are simulated. A target cell bound λ is delivered by choosing the atom's realized rate to invert (6), after which the true $1 - \bar{\pi}$ at each rung follows by aggregation.

The first pass covered the boundary at which every respondent is rationalizable, two interior designs, a design in which a single cell carries the entire shortfall, and a design in which many atoms sit exactly at the non-differentiable point. Minimum coverage was 0.983, taken across all

five rungs. At the boundary design the bound never exceeded zero.

The stress test extends this in two directions. Designs are calibrated by bisection so the true shortfall at the finest rung equals 0.050, 0.065 and 0.080, covering and bracketing the empirical estimate. Within-household outcome dependence is induced by a shared-uniform copula, in which member i of household h uses the household draw U_h with probability ρ and an independent draw otherwise, which leaves every atom's marginal probability exactly unchanged while generating positive within-household dependence. The parameter ρ is the mixing probability of that construction, not a Pearson correlation, and the induced outcome correlation depends on the atom's marginal. Each of the nine combinations uses 1,000 Monte Carlo replications. The minimum coverage anywhere in this grid is 0.990 against a nominal 0.95, above the 0.983 of the first pass, and dependence up to $\rho = 0.7$ does not reduce it.

C Belief Measurement

C.1 Focal responses and the grid rule

Subjective probabilities in the Health and Retirement Study are heaped. Among respondents in the estimation sample the three largest exact response combinations are a survival belief of one half, of one, and of zero. A respondent who reports zero and is treated as holding the point belief zero must, under rational expectations, place no mass whatever on the event, so a single occurrence in that cell is unrationalizable. Symmetrically a report of one hundred admits no failures. Because heaping concentrates at exactly these points, the measurement model determines how much of the identified shortfall is attributable to reporting behaviour.

The classification rule is fixed before estimation and uses every subjective-probability question the respondent answered at the anchor wave, on average 6.0 questions. Let A_i denote the set of answers. The reporting grid is

$$g_i = \begin{cases} 1 & \text{if some } x \in A_i \text{ is not a multiple of 5,} \\ 5 & \text{else if some } x \in A_i \text{ is a multiple of 5 but not of 10,} \\ 10 & \text{else if some } x \in A_i \text{ is a multiple of 10 and not in } \{0, 50, 100\}, \\ 50 & \text{otherwise.} \end{cases}$$

The finest granularity a respondent ever displays reveals the grid on which they are answering, so coarseness is inferred from the respondent's own pattern. Table 1 reports the distribution.

A report x_i from a respondent with grid g_i becomes $[\, x_i - g_i/2,\ x_i + g_i/2 \,]$ truncated to $[0, 1]$. This is the interval within which the underlying point belief is taken to lie, and it enters the bound through Proposition 4. A zero from a grid-one respondent remains essentially zero. A zero from a grid-ten respondent becomes $[0, 0.05]$. A zero from a focal-only respondent would become $[0, 0.25]$, but focal-only respondents are excluded from the authoritative specification.

These intervals are respondent-specific and inferred from respondent behaviour. Widening a

cell to the edges of an analyst-chosen partition, and then permitting the program to select the most favourable belief inside it, would model the analyst's binning choice as though it were the respondent's own uncertainty. Those are different objects and only the second is a measurement model.

C.2 Rounding versus genuine imprecision

The grid rule identifies how coarsely a respondent reports. It does not identify whether a focal-only respondent holds a precise point belief and rounds it, or holds a genuinely imprecise belief that no point summarizes. Direct probing of probabilities about late-onset dementia finds both types, in roughly equal proportions (Giustinelli et al., 2022). For the second type the benchmark $p = \mathbb{P}(Y \mid I)$ is not well defined, so the point-belief version of rational expectations has no testable content, and counting such respondents as rationalizable by default would inflate $\bar{\pi}$ for a reason unrelated to their beliefs.

The authoritative specification therefore removes focal-only respondents from point-belief inference and reports them separately. Of the 16,780 respondents who report a survival belief and satisfy every other sample criterion, 3,218, or 19.2 percent, report only on the focal grid. Removing them leaves the 13,562 respondents of the estimation sample.

C.3 Precise reporters

If the movement between the literal and interval specifications were an artifact of widening intervals, then restricting to respondents who report on a fine grid, with no widening at all, should not reproduce it. Table 6 shows that it very nearly does. Precise reporters taken literally give a bound of 0.910 against 0.918 when every respondent is given their own interval. Two specifications with opposite mechanics agreeing indicates that precise and coarse reporters differ genuinely, which is consistent with evidence that, among Finnish men, the share who report round inflation forecasts falls with measured IQ (D'Acunto et al., 2023).

D Data Construction

All variables come from the RAND HRS Longitudinal File, which places the core interviews on harmonized definitions across waves (Health and Retirement Study, 2025; RAND Center for the Study of Aging, 2025).

D.1 Beliefs and realization

The survival belief is the reported percent chance of living to a target age, divided by one hundred. The instrument adapts that target to the respondent's current age and records it, so the event being forecast differs across respondents and the realization has to be constructed individually.

The reporting-grid classification of Appendix C uses every subjective-probability item the anchor wave contains. These are the chances of living to the adapted target age and to age seventy-five, of entering a nursing home within five years, of working after age sixty-two and after age sixty-five, of leaving a bequest of at least ten thousand dollars, of leaving one of at least one hundred thousand dollars, of leaving any bequest at all, and of receiving an inheritance. Responses outside $[0, 100]$ are treated as missing.

Let T_i denote the target age and let A_i denote age at death, which is observed for decedents. Let $\bar{a}_i$ be the oldest age at which the respondent is observed alive at an interview between waves 8 and 16. The realization is

$$Y_i = \begin{cases} \mathbf{1}\{A_i \geq T_i\} & \text{if } A_i \text{ is observed,} \\ 1 & \text{if } A_i \text{ is missing and } \bar{a}_i \geq T_i, \\ \text{unresolved} & \text{otherwise.} \end{cases}$$

The third case is a respondent alive at last observation who has not yet reached the target age. Such respondents are retained and their mass is allocated as in Proposition 5. They are 38.1 percent of the estimation sample.

D.2 Conditioning variables and sample

The ladder conditions on respondent sex, on age group at the anchor wave, coded as under seventy, seventy to seventy-nine, and eighty or above, on an indicator for fair or poor self-rated health, and on an indicator for current smoking. All four are attributes the respondent unambiguously knows about themselves at the time of the interview, which is the criterion for admitting a variable to the asserted information set.

The estimation sample consists of respondents interviewed at the anchor wave with a non-missing survival belief in $[0, 1]$, a classifiable reporting grid, non-missing sex and age group, and non-missing design identifiers. Respondents who report only on the focal grid are excluded from the authoritative specification and reported separately. This yields 13,562 respondents, of whom 12,745 carry a positive respondent weight.

No respondent is dropped for having an unresolved realization, since that would select on the outcome. Every rung is evaluated on the same 13,562 respondents. Missing sex, age group or design identifiers remove a respondent from the sample entirely, before any rung is computed, and the two health variables are coded as indicators that take the value zero when the underlying item is missing, so that no respondent enters one rung and leaves another. Six respondents have no self-rated health and ninety-five have no smoking status, and they are treated as not in fair or poor health and not currently smoking. Holding the sample fixed allows the decline along the ladder to be read without composition changes, as the nesting in Theorem 1 requires. Respondents with zero or missing survey weight are excluded from weighted quantities by construction.

D.3 Cells

The finest partition consists of 244 atoms defined by the interaction of a coarse belief group with all four conditioning variables. A cell's belief interval is the union of its members' own rounding intervals, which contains each member's interval and therefore preserves validity of the bound while avoiding an analyst-chosen band. Coarser rungs are obtained by aggregating atoms and never by re-partitioning, which makes the ladder nested by construction.

E Survey Design and Weighting

E.1 Design identifiers and estimand

The Health and Retirement Study is a stratified multistage sample (Heeringa and Connor, 1995), and the study supplies sampling-error computation codes for design-based variance estimation, a standard-error stratum and, within it, a half-sample (RAND Center for the Study of Aging, 2025). In the estimation sample these give 56 strata with exactly two half-samples each, which is the ultimate-cluster configuration those codes are constructed for. Estimation uses the person-level analysis weight for the anchor wave.

Weighted quantities describe the population represented by positive values of that weight at Wave 7, which is community-dwelling older Americans in 2004, and not the United States population. Two exclusions follow from the weight's construction. Age-ineligible younger spouses carry zero weight, and in this sample they are 817 respondents with mean age 47.7 against 65.5 among the weighted. Institutionalized respondents are represented by a separate nursing-home weight. The difference between the weighted and unweighted columns of Table 3 reflects the first of them.

A third restriction is larger than either. The estimation sample excludes the 3,218 respondents, 19.2 percent of those otherwise eligible, who report only on the focal grid. The weighted estimand is therefore the community-dwelling older population of 2004 among respondents who report on a finer grid. Reweighting to recover the full population is not possible, because the excluded group is defined by the absence of the object being bounded.

E.2 Replication and linearization

Balanced repeated replication uses a Sylvester Hadamard matrix of order 64 over the 56 strata. In replicate r the half-sample selected by the sign pattern receives twice its analysis weight and the complementary half-sample receives zero, and the variance follows from (12). The number of replicates is the smallest power of two exceeding the number of strata, and the first 56 columns of the matrix are used.

For a stratified design with two units per stratum the linearized variance of a weighted mean reduces to $\sum_h (u_{h1} - u_{h2})^2$, where $u_{hi} = \sum_{j \in hi} w_j (y_j - \hat{\theta}) / \sum_j w_j$ is the unit total of the linearized variable. Panel A of Table 4 compares this with the replication estimator on four smooth statistics.

The ratio is 1.00 in every case, which validates the replicate construction. Appendix B assesses coverage for the non-smooth bound by simulation.

E.3 Clustering beyond the household

The exact test assumes independence within a cell. Household dependence is the dominant violation, since 53.0 percent of the estimation sample share a household with another sample member and more than half the cells contain such a pair. Appendix F reports the test with one respondent per household and under a household-clustered bootstrap. Table 2 also reports a design-aware route in which the same inequalities are evaluated with balanced-repeated-replication standard errors over the strata and half-samples, so that dependence is permitted at the level of the sampling-error half-samples, which nest households. That route replaces the binomial with a normal approximation.

F Additional Robustness

F.1 Measurement and weighting

Table 6 reports the bound under literal reports, under a restriction to respondents reporting on a grid of five or finer with no interval widening, and under respondent-specific rounding intervals. Literal reports give the tightest bound because a reported zero or one hundred then imposes an absolute restriction. The near-agreement between the precise-reporter and interval specifications, obtained through opposite mechanics, indicates that the difference from the literal specification reflects heterogeneity between precise and coarse reporters. Inference is reported for the authoritative specification only, because the design-based procedure was assessed for that specification.

The table is computed on the subsample of respondents who report beliefs in both survival and nursing-home entry, so that the three measurement specifications are compared on a fixed sample and the second domain can be shown to behave the same way. Its levels are therefore not comparable with the survival estimates in the main text, which use the full estimation sample.

The zero-weight group behind the difference between the weighted and unweighted ladders is described in Appendix E. The ordering across rungs is the same in the two columns, with age producing the largest single change and health and smoking adding further. Sex is the one rung where the columns differ in kind. It leaves the weighted bound unchanged at 0.9776, while the unweighted bound moves from 0.9642 to 0.9621, a decline of about two tenths of a percentage point. The exact null response to sex is a property of the weighted estimand and does not carry over to the unweighted one.

F.2 Household dependence

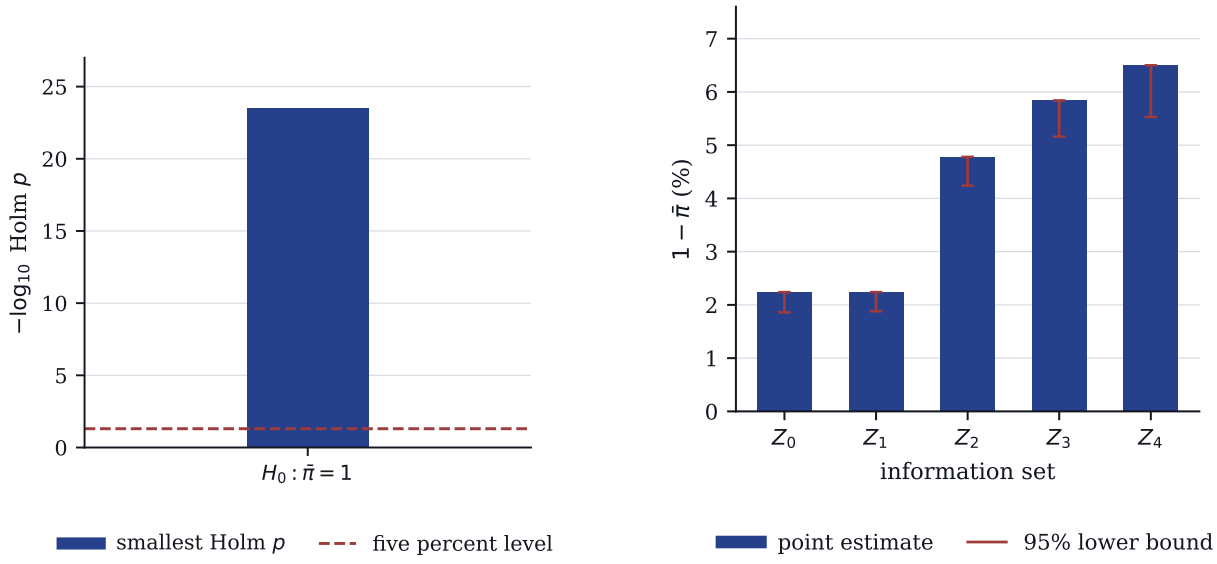
Table 2 reports the exact test under four treatments. Retaining one respondent per household, selected by a deterministic identifier rule that does not use the outcome, leaves the smallest Holm-adjusted p-value at 3.2×10^{-51}, which is smaller than the full-sample value, so the rejection is not an

Table 6: Sensitivity of the bound to the belief-measurement model

Measurement specification	Survival	Second domain	Status of focal-only reporters
Literal reports	0.861	0.938	retained at face value
Precise reporters only, grid ≤ 5	0.910	0.969	excluded
Respondent-specific intervals	0.918	0.982	retained with own interval

Notes. Point estimates of $\bar{\pi}$, unweighted, computed on the subsample holding beliefs in both domains so that the three measurement specifications in the rows are comparable to one another. The second domain is nursing-home entry within five years and is retained here only to show that the measurement model, not the domain, drives the sensitivity. Inference is not reported for these specifications because the design-based procedure was validated for the authoritative specification only, and reporting a bound without a validated confidence statement would invite the two to be read together. Taking reports literally imposes an absolute restriction on respondents who answer zero or one hundred and therefore yields the tightest bound. Restricting to respondents who report on a grid finer than multiples of ten, with no interval widening at all, gives almost the same answer as giving every respondent their own rounding interval, which indicates that the movement reflects a genuine difference between precise and coarse reporters and not the mechanical effect of widening intervals.

artifact of counting spouses twice. A bootstrap that resamples households leaves it at 3.2×10^{-24}, and the half-sample replication route, which permits dependence at the level of the sampling-error half-samples, leaves it at 1.9×10^{-30}. Household and design dependence cost orders of magnitude and leave the conclusion unchanged. Figure 6 places that rejection beside the identified magnitude.



(a) Existence

(b) Magnitude

Figure 6: Strength of the rejection against the identified magnitude

Notes. Panel (a) reports the negative base-ten logarithm of the smallest Holm-adjusted p-value for the hypothesis $\bar{\pi} = 1$ under the household-clustered treatment, the most conservative of the three routes in Table 2. The dashed reference line is the five percent level. Panel (b) reports the weighted point estimate of $1 - \bar{\pi}$ in percent at each rung of the ladder, with whiskers running down to the one-sided 95 percent lower confidence bound from balanced repeated replication. The two panels are on unrelated scales.

F.3 Finite-sample benchmark

Refining cells lowers a plug-in bound mechanically, because $\bar{\lambda}$ is concave in the cell mean and sampling noise therefore biases the plug-in downward, with the bias growing as cells shrink. Under literal beliefs the effect is large. A population that is rational by construction, evaluated on cells of realistic size with realistic base rates, returns a benchmark that falls from roughly 0.97 with a small number of cells to roughly 0.72 once several conditioning variables are added. A ladder computed under literal beliefs is therefore uninterpretable, since most of what it displays would appear in a population that is rational by construction. The interval treatment is authoritative because its benchmark is small.

Under the authoritative specification the belief intervals convert the kink of Proposition 2 into the plateau of Proposition 4. On a plateau the bound is locally constant in the cell mean, so the first-order bias vanishes. A cell mean can still leave the plateau when the cell is small. Simulating a rational population in which each respondent's own reported belief generates their outcome, holding the censoring pattern fixed, gives a benchmark of 0.00 percentage points at Z_0 and Z_1, 0.20 at Z_2, 0.55 at Z_3 and 1.04 at Z_4. The benchmark is an order of magnitude below the literal-belief one and roughly a sixth of the estimate at the finest rung, which is why the interval specification is the one in which the ladder can be read.

That benchmark is also the harsher of two nulls. The coverage designs of Table 4 place each cell's success probability at the midpoint of its belief interval, where the plateau is widest and the bound is least sensitive to sampling noise. Generating outcomes from each respondent's own belief instead puts the cell mean wherever the members' beliefs put it, sometimes near an interval edge. The own-belief null is therefore the least favourable of the two, and the benchmark reported here comes from it.

F.4 The exact lower bound

The exact route of Appendix B, which places simultaneous Clopper–Pearson intervals on every cell proportion and weight, yields a lower bound on $1 - \bar{\pi}$ of roughly a fifth of the point estimate at the base rung. Its validity requires no asymptotics, and it holds under independent sampling for the unweighted empirical distribution. Its conservatism is driven by the multiplicity correction.

F.5 Ordering, sharpness and concentration

Table 7 places the authoritative ladder beside the implementation in which the respondent's belief interval enters the partition key. The two differ only in how cells are formed. The authoritative partition groups respondents by a coarse belief group and assigns the cell the union of their intervals, which lets the cell maximisation choose a belief no member admits. The sharp partition keys on the interval itself, so every cell is homogeneous in $[a, b]$ and the union step disappears. Coarser rungs still aggregate the same atoms in both, so each ladder is nested by construction and monotonicity holds exactly.

The sharp partition is closer to the identification result and worse as an estimator. Removing the union relaxation can only tighten the population bound, so the sharp column would be the object to report if it could be estimated, and the cells that make it sharp are too small to support the plug-in.

Table 7: The ladder under the conservative and the sharp implementation

Rung	Information set Z_j	Conservative	Sharp
Z_0	belief only	2.24	3.20
Z_1	+ own sex	2.24	3.39
Z_2	+ own age group	4.78	6.65
Z_3	+ own self-rated health	5.84	8.04
Z_4	+ own current smoking	6.50	9.22

Notes. Weighted minimum non-rationalizable share $1 - \bar{\pi}$ in percent. The conservative column is the authoritative specification of Table 3, built on 244 atoms whose belief interval is the union of their members' intervals. The sharp column keys the partition on the respondent's own interval and uses 889 atoms, which removes the union relaxation. The sharp column is not usable as an estimate. Its atoms hold a median of 4 respondents against 20 in the conservative partition, with 67.5 percent holding fewer than ten, and at that cell size the plug-in bound is badly biased. Under a rational population its benchmark reaches 4.54 percentage points at the finest rung, about half of the 9.22 reported, and the one-sided lower bound covers in 0.42 of replications in the design battery and 0.27 in the stress grid, against a nominal 0.95. The conservative partition covers at 0.983 and above throughout.

The increments in Table 3 are order-specific, since each rung conditions on the previous one. Averaging a variable's marginal contribution over all orderings of the four variables gives the Shapley decomposition in Figure 7, whose components sum by construction to the total rise of 4.25 percentage points.

Figure 7 shows that age group dominates at any position, so the ranking in the main text is a property of the data. Every whisker has positive width, so every contribution moves with position. Sex is the only one that can vanish, and its whisker still reaches 0.74 points, so the exact null in Table 3 is a statement about the ordered ladder.

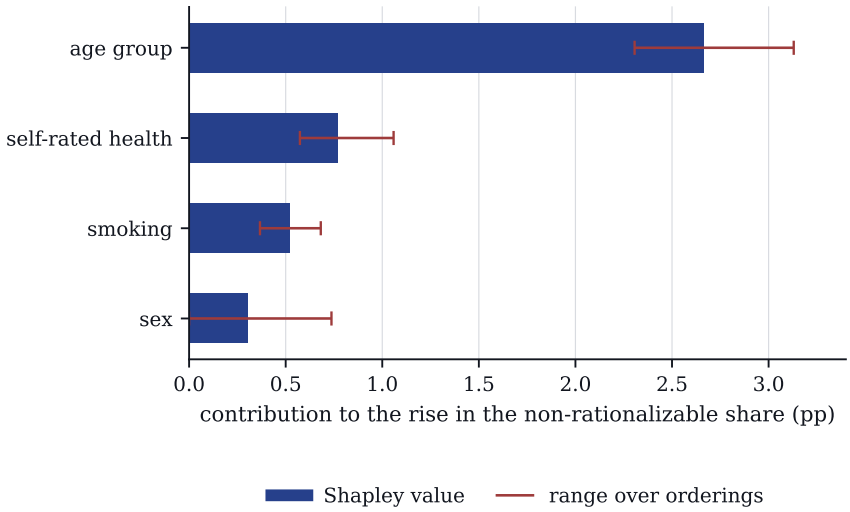


Figure 7: Contribution of each conditioning variable across all orderings

Notes. Bars are Shapley values, the average marginal contribution of each variable to the rise in the weighted minimum non-rationalizable share over the twenty-four orderings of the four conditioning variables, in percentage points. Whiskers run from the smallest to the largest marginal contribution the variable attains at any position. The four bars sum to the total rise of 4.25 points. Every subset bound is computed on the same sample and the same finest partition.

Figure 8 turns to where the shortfall comes from. Each cell contributes $w_c(1 - \bar{\lambda}_c)$ to the total, and ordering cells from the largest contribution downward traces the cumulative share of the shortfall against the cumulative share of the weighted population.

The two curves differ in the size of their steps. Without conditioning, the shortfall is the work of a few cells at the tails of the belief distribution, the largest carrying 47.1 percent of it, and dropping the five largest contributors leaves 0.00 percentage points. With the full information set, 160 of 244 cells contribute, the largest carries 3.8 percent, and dropping the ten largest still leaves 4.70 of 6.50 points. What conditioning uncovers is dispersed across many cells.

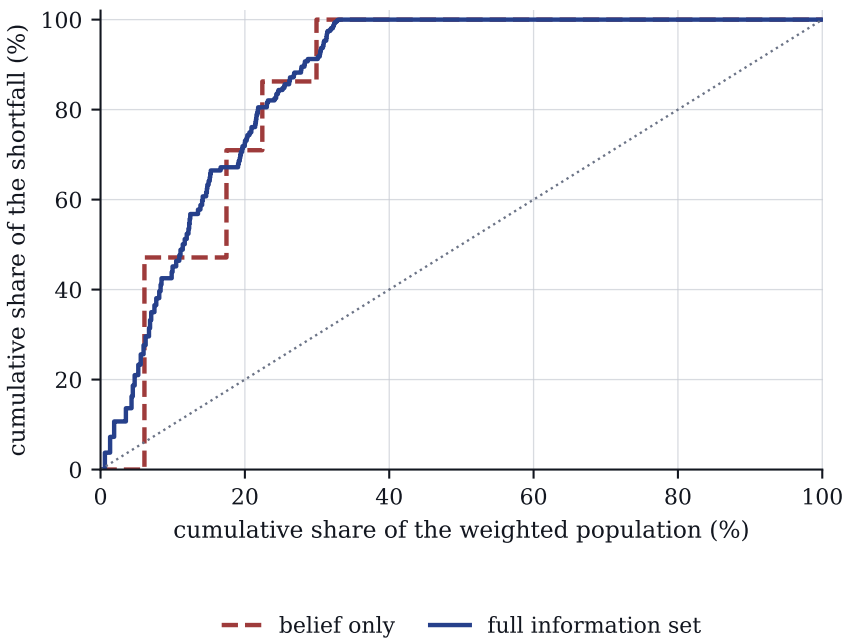


Figure 8: Concentration of the shortfall across cells

Notes. Cells are ordered by their contribution $w_c(1 - \bar{\lambda}_c)$ to the weighted minimum non-rationalizable share, largest first. The horizontal axis is the cumulative share of the weighted population, the vertical axis the cumulative share of the total shortfall. The dotted line marks proportionality, where a given share of the population accounts for the same share of the shortfall. Both curves flatten near a third of the population, since only cells with $\bar{\lambda}_c < 1$ contribute at all. They differ in the size of the steps. The base rung curve climbs in 4 jumps, the first of which is 47.1 percent of the total, while the finest rung curve climbs in 160 small ones.

F.6 Where the shortfall sits

Figure 8 shows the shortfall at the finest rung is spread across many cells. Which dimension organizes that spread is a separate question, and the finest partition answers it directly. Each cell belongs to exactly one category of each conditioning variable, so summing $w_c(1 - \bar{\lambda}_c)$ within a category decomposes the total exactly, once per dimension. The object is a group's contribution to a population bound.

Age is by far the dominant organizing dimension. Respondents aged seventy to seventy-nine are 17.0 percent of the weighted population and account for 51.2 percent of the shortfall. Those eighty or older are 6.5 percent and account for 28.7. Together, under a quarter of the population carries about four fifths of it, while the 76.6 percent under seventy carry 20.1 percent. Fair or poor health is mildly disproportionate, 22.8 percent of the population against 38.5 of the shortfall. Smoking and sex are close to proportional. Across belief bands the contributions track population shares except at the top, where the 6.1 percent reporting a survival chance above ninety-five carry 16.3 percent.

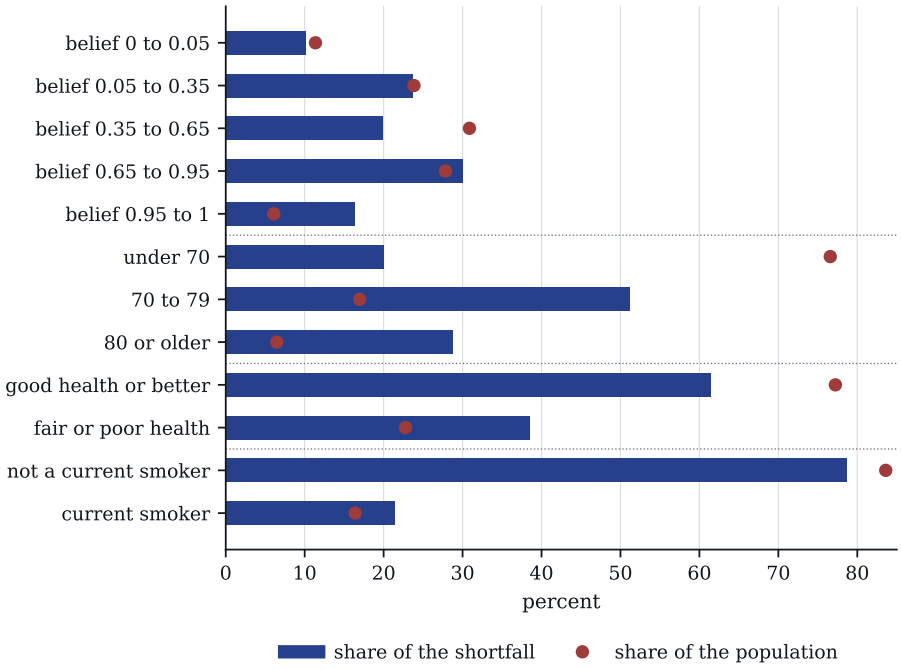


Figure 9: Contribution to the shortfall by group

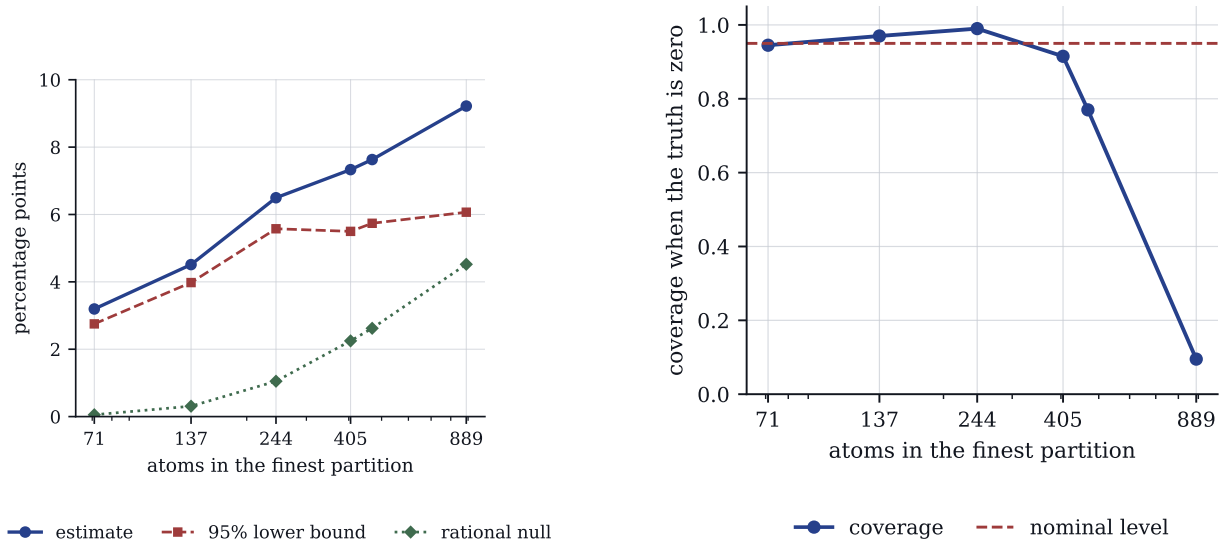
Notes. Bars are each group's share of the total weighted shortfall at the finest rung, computed as the sum of $w_c(1-\bar{\lambda}_c)$ over the cells belonging to that group. Circles are the group's share of the weighted population. A bar exceeding its circle means the group contributes more than its size. Dotted lines separate dimensions, and within each dimension the bars sum to one hundred. Sex is omitted from the figure because both categories are within three points of proportional. The rational-null benchmark is below 0.6 percentage points for every group, so it does not reorder them.

The decomposition says something the ladder does not. The mass that cannot be rationalized

sits with the oldest respondents.

F.7 The identification-precision frontier

The conservative and sharp partitions of Table 7 are two points on a curve. Grouping the reported belief to the nearest half, fifth, tenth, twentieth or fiftieth, and finally keying on the respondent's own interval, gives a family of partitions running from 71 to 889 atoms. Figure 10 traces the estimate, its lower bound, the rational-null benchmark and the coverage of the one-sided bound when the truth is zero, which is the case where a false positive would be most damaging.



(a) Identification

(b) Precision

Figure 10: Refining the partition tightens the bound and weakens the inference

Notes. Each point is one partition of the belief, from the nearest half to the respondent's own interval, always interacted with the same four conditioning variables and evaluated at the finest information set. Panel (a) reports the weighted estimate of $1 - \bar{\pi}$, its one-sided 95 percent lower bound, and the shortfall a population that is rational by construction produces on the same cells. Panel (b) reports the share of simulations in which the lower bound stays below the truth when every respondent is rational, against the nominal level. The horizontal axis is on a log scale.

Both halves behave as the theory predicts, and the second is the binding one. The estimate rises monotonically with refinement, from 3.19 percent at 71 atoms to 9.22 at 889, because each refinement removes some of the union relaxation. The rational-null benchmark rises with it, and faster in proportion. Coverage is 0.945 at the coarsest partition, reaches 0.990 at 244 atoms, and then falls away, to 0.915 at 405, 0.770 at 469 and 0.095 at the respondent's own interval.

The partition used throughout the paper is the finest that retains nominal coverage at the boundary. Finer partitions tighten the bound and lose inferential reliability. The partition was fixed in advance as the grouping of beliefs to the nearest tenth and the curve was computed afterwards. The curve also traces the reliability of the inference without reference to the estimate, since coverage under a rational null requires no knowledge of the true shortfall.

F.8 What the conventional regression sees

The classical way to test efficiency is to regress the outcome on the reported belief and the asserted information,

$$Y_i = \alpha + \beta\, p_i + \gamma' Z_i + \varepsilon_i, \tag{15}$$

and to read rational expectations as $\alpha = 0$, $\beta = 1$ and $\gamma = 0$. This treats the report as an exact point belief, so the exercise is a literal-report diagnostic. Censoring has to be respected.

The coefficient is linear in the outcome vector,

$$\hat{\beta}_j \;=\; \sum_i a_{ji}\, Y_i, \qquad a_j \;=\; \text{the } j\text{th row of } (X'WX)^{-1}X'W,$$

and the weights attached to unresolved observations differ in sign. Setting every unresolved outcome to zero and then to one therefore does not bracket the coefficient. It evaluates two arbitrary points inside the feasible set. The sharp interval assigns each unresolved outcome by the sign of its own weight, one to the positive weights and zero to the negative ones to maximize and the reverse to minimize. Since the free outcomes are binary, enter linearly and are unconstrained, those assignments attain the extrema exactly.

Table 8: Forecast-efficiency coefficients, sharp bounds under censoring

	Survival to target (35.1% unresolved)	Work after 62 (9.3% unresolved)
Reported belief p	$[-0.566,\ 0.801]$	$[0.295,\ 0.522]$
Female	$[-0.430,\ 0.519]$	$[-0.140,\ 0.058]$
Second age group	$[-0.337,\ 0.334]$	$[-0.092,\ 0.097]$
Third age group	$[-0.622,\ 0.052]$	$[-0.074,\ 0.091]$
Fair or poor health	$[-0.553,\ 0.261]$	$[-0.235,\ -0.042]$
Current smoker	$[-0.634,\ 0.370]$	$[-0.145,\ 0.052]$

Notes. Weighted least squares of equation (15). Each entry is the sharp interval over all assignments of the unresolved outcomes, obtained by assigning each one according to the sign of its influence weight on that coefficient. Age groups are the second and third categories of each sample's own grouping. Rational expectations relative to the asserted information set implies a coefficient of one on the reported belief and zero on everything else.

The conventional test identifies almost nothing here. In the survival domain, where a third of outcomes are unresolved, no coefficient excludes zero, and the interval on the reported belief runs from -0.566 to 0.801, which contains both zero and one. The regression cannot reject $\beta = 1$, cannot establish it, and cannot sign a single covariate. In the work domain, where under a tenth is unresolved, two survive. The coefficient on the reported belief lies in $[0.295, 0.522]$, which excludes both zero and one, and fair or poor health lies in $[-0.235, -0.042]$, which excludes zero. Sex and smoking do not.

On the same survival data the bound framework delivers 2.24 percent at the base rung and 6.50 at the finest, with lower bounds that have been checked for coverage, while the efficiency regression delivers no signed coefficient at all. The regression needs a conditional mean, and unre-

solved outcomes destroy it. The bound needs only whether the reachable range of outcomes meets the admissible belief interval, and unresolved outcomes widen that range instead of destroying it. Censoring costs the regression its identification and costs the bound some of its power.

G Joint Rationalizability Across Domains

The framework extends to several outcomes elicited from the same individuals in a single block, which raises a question the single-domain bound cannot address. A large share of a population may be rationalizable in each domain separately while a much smaller share is rationalizable in all of them at once. This appendix develops the object, characterizes it for two domains, identifies the data configurations under which it carries information beyond the single-domain bounds, and applies it to the domain pairs the study elicits.

G.1 The joint program

With D binary domains, let $y \in \{0,1\}^D$ index outcome configurations and let g_y denote the observed mass on configuration y within a cell defined by the vector of reported beliefs $(p_1, \ldots, p_D)$ and covariates. Write r_y for the mass of the rational subpopulation at configuration y. Working directly in these masses, instead of in a share times a conditional law, keeps the problem linear:

$$\max_{r \geq 0} \sum_y r_y \quad \text{subject to} \quad r_y \leq g_y \;\; \forall y, \qquad \sum_y y_d \, r_y \;=\; p_d \sum_y r_y \;\; \forall d. \tag{16}$$

Writing the program as a share λ multiplying a free joint law q with marginals p_d would make the marginal constraint bilinear in (λ, q), and the substitution $r = \lambda q$ removes the product. Every coefficient in (16) is then either an indicator or a reported belief, so the program has known coefficients and an estimated right-hand side.

Three extensions preserve linearity. Interval beliefs replace the equality by $a_d \sum_y r_y \leq \sum_y y_d r_y \leq b_d \sum_y r_y$. Censoring enters exactly as in Proposition 5, through allocation variables $\alpha_{g,y} \geq 0$ for each group g of respondents with a partially known configuration, with $\sum_y \alpha_{g,y}$ equal to that group's mass, $\alpha_{g,y} = 0$ for configurations incompatible with what is known, and the componentwise constraint replaced by $r_y \leq \sum_g \alpha_{g,y}$. Weights enter through the right-hand side.

A natural benchmark asks how large the jointly rational share could be if the analyst preserved each domain's observed marginal outcome distribution but were free to choose the cross-domain dependence.

Definition 2 (Marginal envelope)**.** $\pi_{\text{marg}} \;=\; \mathbb{E}\Big[\min_d \bar{\lambda}_d(X)\Big]$, where $\bar{\lambda}_d$ is the single-domain cell bound of Section 3.

Proposition 8 (Envelope)**.** π_{marg} *is exactly the largest jointly rational share attainable while pre-*

serving each domain's marginal outcome distribution and choosing the coupling freely. Consequently

$$\pi_{\text{joint}} \;\le\; \pi_{\text{marg}} \;\le\; \min_{d} \bar{\pi}_d. \tag{17}$$

The proof is at the end of this appendix. The decomposition

$$\underbrace{1 - \pi_{\text{marg}}}_{\text{marginal inconsistency}} \qquad \text{and} \qquad \underbrace{\pi_{\text{marg}} - \pi_{\text{joint}}}_{\text{additional joint restriction}} \tag{18}$$

separates what the marginals already imply from what the observed dependence adds.

G.2 Two domains

For $D = 2$ the program admits a closed form, so no solver is required.

Proposition 9 (Two-domain characterization)**.** *Let $\delta = p_1 + p_2 - 1$ and let $(g_{00}, g_{01}, g_{10}, g_{11})$ be the observed cell masses. Of the sixteen pairwise feasibility conditions implied by (16), twelve are vacuous or reproduce the four inequalities defining $\bar{\lambda}_1$ and $\bar{\lambda}_2$. The remaining four are*

$$S \le \frac{g_{11}}{p_1 + p_2 - 1} \qquad \text{if } p_1 + p_2 > 1, \tag{19}$$

$$S \le \frac{g_{00}}{1 - p_1 - p_2} \qquad \text{if } p_1 + p_2 < 1, \tag{20}$$

$$S \le \frac{g_{10}}{p_1 - p_2} \qquad \text{if } p_1 > p_2, \tag{21}$$

$$S \le \frac{g_{01}}{p_2 - p_1} \qquad \text{if } p_2 > p_1, \tag{22}$$

and the cell's jointly rational mass is

$$S^*(x) \;=\; \min\Big\{\bar{\lambda}_1,\; \bar{\lambda}_2,\; 1,\; \textit{whichever of (19) to (22) apply}\Big\}.$$

Writing $S^{\text{marg}}(x) = \min\{\bar{\lambda}_1, \bar{\lambda}_2, 1\}$ for the cell-level envelope, $S^*(x) = S^{\text{marg}}(x)$ if and only if every applicable constraint among (19) to (22) is slack at $S^{\text{marg}}(x)$. The population objects of Definition 2 and Proposition 8 are the expectations of these cell quantities over the distribution of cells, so $\pi_{\text{joint}} = \mathbb{E}[S^*(X)]$ and $\pi_{\text{marg}} = \mathbb{E}[S^{\text{marg}}(X)]$.

The reduction was verified numerically against a linear-programming solution on four hundred randomly generated full-support cells with random beliefs, with a maximum absolute discrepancy of order 10^{-16}.

The four constraints are Fréchet restrictions. A rational subpopulation whose two marginals are p_1 and p_2 must place at least $S(p_1 + p_2 - 1)$ of its mass on joint occurrence, at least $S(1 - p_1 - p_2)$ on joint non-occurrence, and at least $S|p_1 - p_2|$ on the corresponding discordant configuration. The data must supply that mass, and when they do not the joint bound falls well below the envelope.

Example 1. Let $p_1 = p_2 = 0.9$ and $(g_{00}, g_{01}, g_{10}, g_{11}) = (0.11, 0.44, 0.44, 0.01)$, so every configuration has positive mass. Both marginals equal 0.45, hence $\bar{\lambda}_1 = \bar{\lambda}_2 = 0.5$ and $\pi_{\text{marg}} = 0.500$. But (19) requires $r_{11} \geq 0.8\,S$ while only $g_{11} = 0.01$ is available, so $\pi_{\text{joint}} \leq 0.0125$. The joint restriction reduces the bound by a factor of forty on data with full support.

Example 1 shows the joint object is not a relabelling of the minimum of the marginal bounds. It also shows where its power lies. Two distinct mechanisms activate the extra constraints. Beliefs far from one half activate the co-occurrence and non-occurrence bounds (19) and (20), since the required joint mass $S(p_1 + p_2 - 1)$ is positive only when $p_1 + p_2 > 1$ and $S(1 - p_1 - p_2)$ only when $p_1 + p_2 < 1$. Beliefs that differ across domains activate the discordance bounds (21) and (22), which require mass $S|p_1 - p_2|$ on the discordant configuration and apply even when both beliefs sit near one half, as at $p_1 = 0.45$ and $p_2 = 0.55$. Whether an applicable restriction actually binds in a given cell is a property of that cell's observed masses, as Proposition 9 states, and the population gap $\pi_{\text{marg}} - \pi_{\text{joint}}$ averages those cell-level shortfalls. What the beliefs govern is which restrictions apply, and none applies when $p_1 + p_2 = 1$ with $p_1 = p_2$, so the joint object is weakest when beliefs are near one half *and* close to one another. For D domains the analogous condition for the all-ones constraint is that the mean belief exceed $(D-1)/D$, which is 0.50, 0.67 and 0.75 at two, three and four domains, so the belief levels required for the joint object to apply rise as domains are added. A set of domains whose beliefs cluster near one half and resemble each other will reproduce the marginal envelope no matter how large the sample.

A simulation illustrates the magnitude of the effect. Constructing populations in which a known share is rational and varying whether the non-rational share fails in all domains simultaneously or independently across them, the joint bound tracks the marginal envelope when beliefs are drawn from a low range, separates partially when they are drawn from a middle range, and separates most when they are drawn from a high range. With three domains, beliefs in the upper range and independent failures, the joint bound returns roughly 0.23 where the marginal envelope returns roughly 0.50 and the true jointly rational share is 0.125. The bound remains an upper bound and moves most of the distance.

G.3 Empirical application

Proposition 9 is a design criterion as much as a characterization. It says to look for domains whose beliefs are dispersed and whose levels differ, so that the co-occurrence and discordance requirements apply to a large share of the population. That criterion involves only the belief distributions, so the pair used here was selected by applying it to those distributions before any joint bound was computed. The study elicits a second personal probability that meets the criterion, the chance of working full time after age sixty-two, asked of respondents who have not yet reached that age.

That belief has a mean near one half and a standard deviation of 0.36, against survival beliefs concentrated well above one half among the same respondents, so the two levels differ in the direction the criterion asks for. Among the 3,397 respondents who report both on a non-focal grid,

49.1 percent of the weighted mass sits in cells where the two beliefs sum to more than one, which is where (19) applies.

The additional restriction is small. Over 55 cells the maximum rationalizable share is 0.9972 for survival and 0.8975 for work, the marginal envelope is 0.8974, and the joint bound is 0.8958. Requiring the same people to be rationalizable in both domains at once therefore costs 0.17 percentage points beyond what the two marginals already imply. A population that is rational by construction returns 0.00 percentage points for the same object on the same cells, so the gap is not a finite-sample artifact, and it is an order of magnitude below the marginal inconsistency it is added to.

The constraints apply widely here, and they bind only where the data fail to supply the required joint mass. In this population the data mostly do supply it, because respondents who report a high survival chance together with a low chance of working are common enough to fill the discordant configurations the Fréchet inequalities demand. The joint object is therefore active and identified and still close to the envelope, which is a statement about how cross-domain belief data behave.

Nursing-home entry is the contrast case, a pair that meets the criterion in almost no cell. Applied to survival and nursing-home entry, the joint bound falls below the marginal envelope by under two percentage points, and essentially the entire gap comes from the single cell in which the two beliefs sum to more than one. Survival beliefs in this population are dispersed but nursing-home beliefs are concentrated near zero, so the co-occurrence restriction is inactive almost everywhere.

The two pairs together make the design lesson concrete. Selecting domains on the position of their belief distributions moves the share of the population facing an applicable constraint from almost nothing to about half. Selecting them on availability alone does not. In neither pair does the applicable constraint bind hard. Example 1 shows the same restriction cutting a bound by a factor of forty on data with full support.

G.4 Proofs

Proof of Proposition 8

Upper direction. Let r solve (16) with total mass S. For each domain d, summing the componentwise constraints over configurations with $y_d = 1$ gives $p_d S = \sum_y y_d r_y \le \sum_{y:y_d=1} g_y = m_d$, and over configurations with $y_d = 0$ gives $(1-p_d)S \le 1 - m_d$. Hence $S \le \bar{\lambda}_d$ for every d and therefore $S \le \min_d \bar{\lambda}_d$.

Attainability. Fix the cell's marginals m_d and set $S = \min_d \bar{\lambda}_d$. Let Q be any joint law on $\{0,1\}^D$ with marginals p_d, which exists, for instance the product law. Let A be any non-negative measure of total mass $1 - S$ with marginals $m_d - S p_d$, which exists because those quantities are non-negative exactly when $S \le m_d/p_d$ and their complements $1 - m_d - S(1-p_d)$ are non-negative exactly when $S \le (1-m_d)/(1-p_d)$, both implied by $S \le \bar{\lambda}_d$. Then $G = S\,Q + A$ has marginals m_d and admits rational mass S. Since the analyst is free to choose the coupling, π_{marg} is attained. The chain (17) follows because $\mathbb{E}[\min_d \bar{\lambda}_d] \le \mathbb{E}[\bar{\lambda}_d] = \bar{\pi}_d$ for each d. □

Proof of Proposition 9

Write $t = r_{11}$. The marginal constraints determine the remaining masses, $r_{10} = Sp_1 - t$, $r_{01} = Sp_2 - t$ and $r_{00} = S(1 - p_1 - p_2) + t$. Non-negativity of the four masses gives

$$t \geq 0, \quad t \leq Sp_1, \quad t \leq Sp_2, \quad t \geq S\delta,$$

and the componentwise constraints $r_y \leq g_y$ give

$$t \leq g_{11}, \quad t \geq Sp_1 - g_{10}, \quad t \geq Sp_2 - g_{01}, \quad t \leq g_{00} + S\delta.$$

Feasibility of S is therefore equivalent to

$$\max\{0,\ S\delta,\ Sp_1 - g_{10},\ Sp_2 - g_{01}\} \ \leq \ \min\{Sp_1,\ Sp_2,\ g_{11},\ g_{00} + S\delta\},$$

which is a system of sixteen pairwise inequalities. Four are vacuous, namely $0 \leq Sp_1$, $0 \leq Sp_2$, $0 \leq g_{11}$ and $S\delta \leq g_{00} + S\delta$. Two hold because $p_d \leq 1$, namely $S\delta \leq Sp_1$ and $S\delta \leq Sp_2$. Two are trivial identities, $Sp_1 - g_{10} \leq Sp_1$ and $Sp_2 - g_{01} \leq Sp_2$. Four reproduce the single-domain constraints, since $Sp_1 - g_{10} \leq g_{11}$ is $S \leq m_1/p_1$, $Sp_2 - g_{01} \leq g_{11}$ is $S \leq m_2/p_2$, $Sp_1 - g_{10} \leq g_{00} + S\delta$ is $S \leq (1-m_2)/(1-p_2)$, and $Sp_2 - g_{01} \leq g_{00} + S\delta$ is $S \leq (1-m_1)/(1-p_1)$. The remaining four are (19) to (22), each active only under the stated sign condition. Hence the cell value is the minimum of $\bar{\lambda}_1$, $\bar{\lambda}_2$, one and the applicable members of (19) to (22), and equality with the envelope holds exactly when all applicable members are slack at $S = \min\{\bar{\lambda}_1, \bar{\lambda}_2, 1\}$. □

Verification of Example 1

With $p_1 = p_2 = 0.9$ and $(g_{00}, g_{01}, g_{10}, g_{11}) = (0.11, 0.44, 0.44, 0.01)$ the marginals are $m_1 = g_{10} + g_{11} = 0.45$ and $m_2 = g_{01} + g_{11} = 0.45$, so $\bar{\lambda}_1 = \bar{\lambda}_2 = \min\{0.45/0.9,\ 0.55/0.1\} = 0.5$ and $\pi_{\text{marg}} = 0.500$. Here $\delta = 0.8 > 0$, so (19) applies and gives $S \leq g_{11}/\delta = 0.01/0.8 = 0.0125$. All four configurations carry positive mass, so the gap is not a consequence of degenerate support. □